Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kilifi County

Final Mwarakaya Ward Adaptation Plan

Consultant

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TABLE OF CONTENTS

List of Figures iii
List of Tables iii

1 Chapter 1: Background information 1
1.1 Introduction 1
1.2 Plan development approach 2
1.2.1 Inception Phase 3
1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA) 4
1.3 Linkages to other planning processes 8

2 Chapter 2: Coordination of Adaptation Actions 9
2.1 Introduction 9
2.2 Current Institutional Arrangements 9
2.3 Relevant Policies 10
2.4 Proposed Coordination Arrangements 10
2.5 Implementation Roles 11

3 Chapter 3: Situation analysis of climate vulnerability 13
3.1 Introduction 13
3.2 Current climate trends 13
3.3 Future climate trends 14
3.4 Climate hazards and vulnerability 16
3.4.1 Drought 16
3.4.2 Pests and Diseases 18
3.4.3 Floods 20
3.4.4 Human wildlife conflicts 22
3.5 Classification of exposure and Resilience through livelihoods 24
3.5.1 Livelihood Structure and Monthly Incomes 25
3.5.2 Vulnerabilities of Livelihood Options 26
3.5.3 Overall Classification and Implications 27
3.6 Analyzing causes of vulnerability. 27
3.6.1 Classification of Severity of the Vulnerabilities and Their Causes 27
3.6.2 Identification of Unsafe Conditions 27
3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability 27
3.6.4 Identification of Underlying Causes of Vulnerability 28
3.6.5 Prioritization of Causes 29

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards 30
4.1 Children's Vulnerability to Climate Hazards and Key Mitigation Measures 33

5 Chapter 5: Adaptation Actions 35
5.1 Introduction 35
5.2 Existing climate-resilient and livelihood-enhancing projects 35
5.3 Adaptation actions 37
5.4 Resource Mobilisation 45

6 Chapter 6: Adaptation Monitoring and Evaluation 46
6.1 Introduction 46
6.2 The Adaptation Theory of Change 46
6.3 Performance Indicators for Adaptation 47
6.4 Adaptation Reporting and Learning 52

7 References 54
8 Acknowledgements 56
9 Annexure 57

List of Figures

Figure 1: Location of Mwarakaya Ward and survey sites (Source: PCVA, 2025) 2
Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014)..... 3
Figure 3: Focus Group Discussions in Mwarakaya Ward (Source: Field Surveys,2025) 6
Figure 4: Hazard and Resource Mapping in Ondoni Village, Mwarakaya Ward (Source: PCVA, 2025) . 7
Figure 5: SSP2 4.5 Rainfall, AET and PET Projections (Source: Chirps database and ERA5-Ag, 2022) ...14
Figure 6: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022) 14
Figure 7: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022) ..15
Figure 8: SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022) 15
Figure 9: Drought prevalence historical trend (Source: PCVA, 2025) 16
Figure 10: Drought vulnerability score in Mwarakaya ward (Source: PCVA, 2025).....16
Figure 11: Kaya Chonyi NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2024).....17
Figure 12: Pests and diseases prevalence historical trend (Source: PCVA, 2025) 19
Figure 13: Pests and diseases vulnerability score in Mwarakaya Ward (Source: PCVA, 2025) 19
Figure 14: Floods prevalence historical trend (Source: PCVA, 2025).....21
Figure 15: Floods vulnerability score in Mwarakaya Ward (Source: PCVA, 2025).....21
Figure 16: Human wildlife conflict prevalence historical trend (Source: PCVA, 2025) 23
Figure 17: Human wildlife conflict vulnerability score in Mwarakaya Ward (Source: PVCA, 2025) 23
Figure 18: Respondents who lacked money to meet basic needs in the last 12 months (Source: PCVA, 2025) 26

List of Tables

Table 1: Institutional Roles in Implementation of Adaptation Actions 12
Table 2: Monthly earnings by livelihood and village (Source: PCVA, 2025) 25
Table 3: Gender/Age Vulnerability Analysis for Climate Hazards 32
Table 4: Existing climate resilient and livelihood enhancing projects in Mwarakaya Ward 36
Table 5: Adaptation Actions for Mwarakaya Ward 40

1 Chapter 1: Background information

1.1 Introduction

Plan international Kenya is implementing the Conservation and Sustainable Management of Marine and Coastal Ecosystems (COSME) Project in Kwale and Kilifi Counties. The two counties are endowed with rich coastal and marine ecosystems ranging from terrestrial forests, mangroves, seagrass beds, coral reefs, marine species and open sea marine resources among others. These natural assets not only underpin livelihoods but also drive key economic sectors including tourism, agriculture, fisheries, and maritime trade. Despite their economic and ecological value, the 2021 country and county climate risk profiles highlight that these resources are increasingly under threat from climate change impacts such as drought, flooding, sea level rise, ocean acidification, rising temperatures, and forest fires. The consequences of these risks exacerbate underlying socio-economic vulnerabilities, with women, youth, and Persons Abled Differently (PAD) experiencing disproportionate impacts due to structural inequalities. Loss of income, displacement, reduced productivity, and damage to infrastructure and markets are already affecting community well-being and resilience in the region.

In response, the COSME project aims to enhance women-led, climate-resilient livelihoods through ecosystem-based approaches. Hence, to support project implementation requires the development of Gender Responsive Adaptation and Preparedness Plans (GRAPPs) that integrate climate risk diagnostics with locally appropriate adaptation pathways. Mwarakaya Ward Climate Change Adaptation Plan (WAP) is therefore prepared as part of the GRAPPs outputs. The ward measures 42.7 km^2, located in Kilifi County at Kenya's southern coast with a population of 16,563 people (7,725 male & 8,838 female) (KNBS, 2019), and faces a high vulnerability to climate change impacts. The ward's climate is a tropical coastal characterized by bimodal rainfall and warm temperatures year-round. It's economy and livelihoods depend heavily on climate-sensitive sectors such as rain-fed agriculture, livestock, and coastal resources (Figure 1). However, recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode development gains and degrade critical natural resources upon which local livelihoods are dependent, making adaptation a critical priority for the ward.

Recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode hard-won development gains, making adaptation a critical priority for the ward. In response, Kenya's policy framework has evolved to support local climate action, from the National Climate Change Response Strategy and Action Plans to the Climate Change Act 2023, which emphasizes devolved climate governance and resilience building at county and community levels. This Ward Adaptation Plan aligns with national and county climate change policies by mainstreaming adaptation into local planning and empowering the community to implement priority resilience measures.

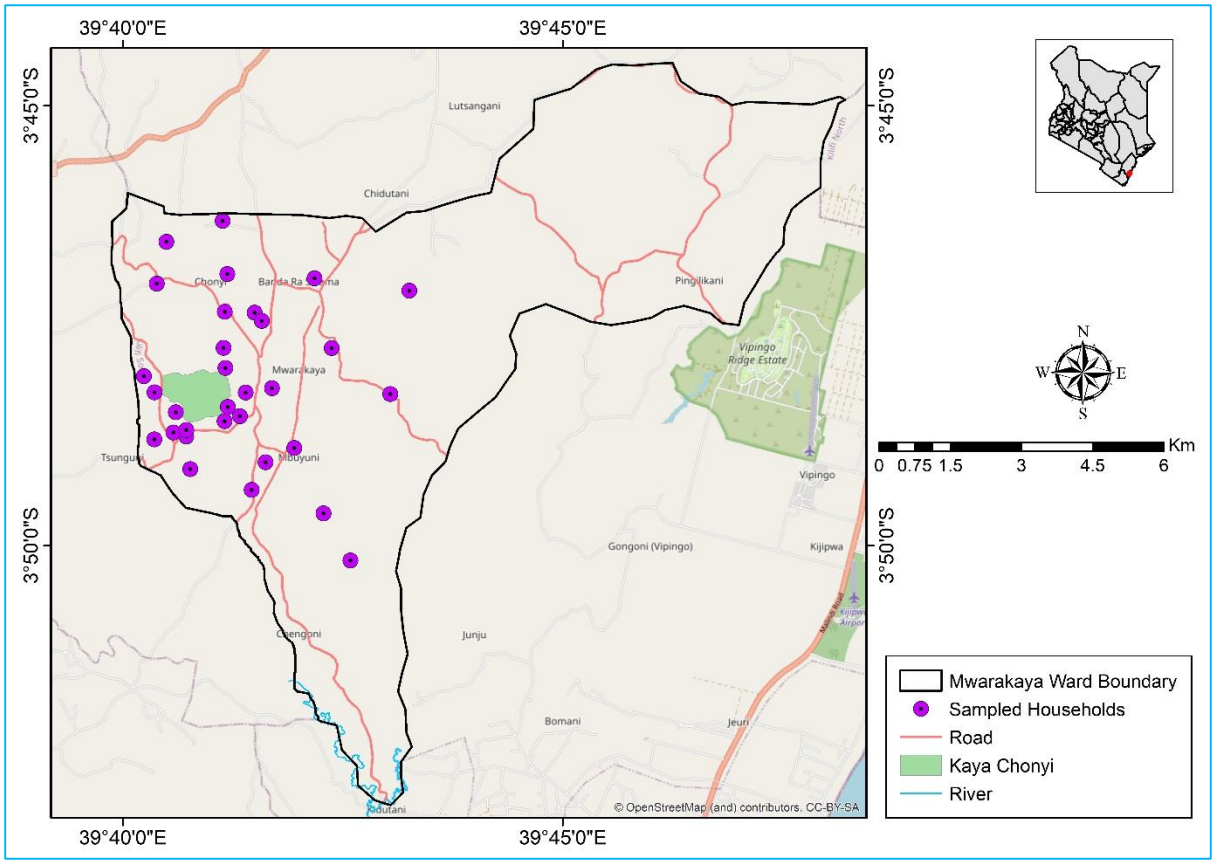

Figure 1: Location of Mwarakaya Ward and survey sites (Source: PCVA, 2025)

1.2 Plan development approach

The Ward Adaptation Plan for Mwarakaya was developed in a participatory, gender-responsive and livelihoods-based vulnerability framework that conceptualized climate risk as an outcome of the interaction between exposure, sensitivity and adaptive capacity (Figure 2). Within this analytical framing, vulnerability was understood as the degree of climate-induced damage experienced by socio-ecological systems, while resilience denoted the system's ability to maintain or rapidly regain functionality following climatic shocks. This approach acknowledged that climate risks in Mwarakaya, characterized predominantly by recurrent droughts, seasonal flooding, pests and diseases and human–wildlife conflicts arising from proximity to Kaya Chonyi, do not affect population groups uniformly. Women, men, youth, elderly persons and Persons Abled Differently (PADs) face differentiated risks due to disparities in livelihood assets, labor burdens, mobility, access to resources, intra-household power dynamics and decision-making autonomy.

The plan was therefore designed using a process that combined experiential community knowledge with quantitative datasets, spatial analysis, ecological diagnostics and climatic modelling. This integrated methodology enabled the plan to link climatic hazards to specific livelihood systems; rainfed agriculture, small-scale trade and casual labour and to examine how these systems are shaped by gender norms, social vulnerabilities and institutional arrangements at ward level. The resulting insights provided a nuanced understanding of how climate extremes intersect with poverty, food insecurity, ecosystem degradation, weak infrastructure and limited institutional support, thereby shaping the context within which adaptation strategies must be developed.

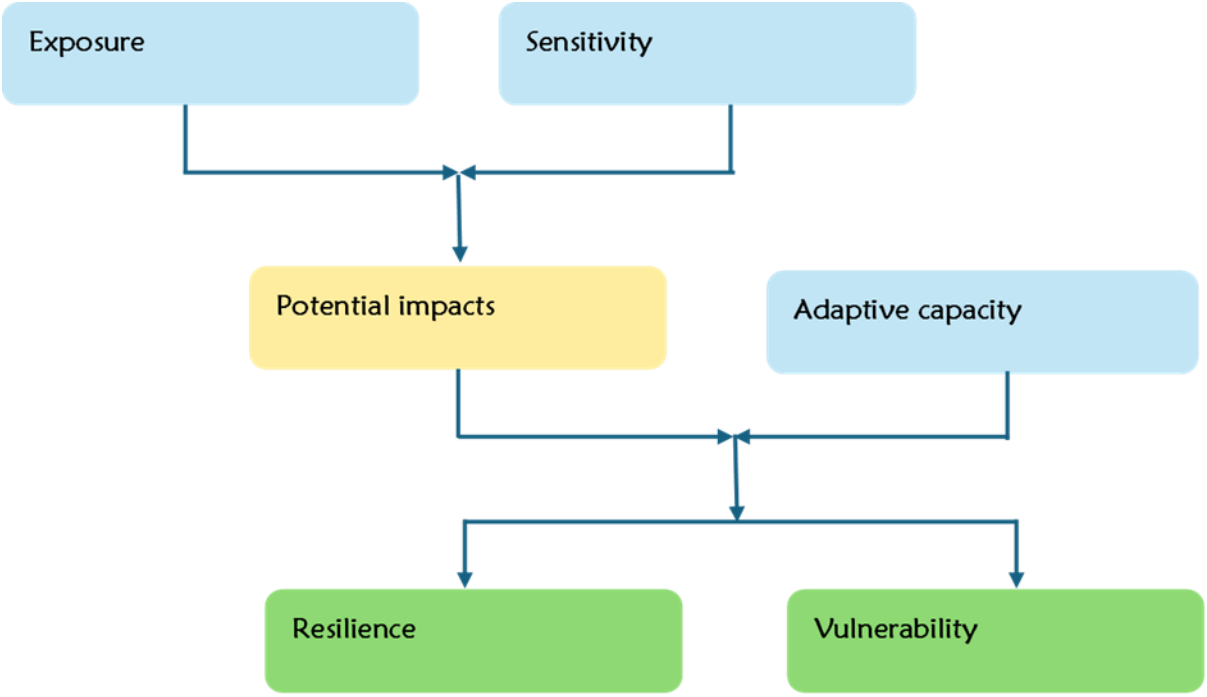

Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014).

1.2.1 Inception Phase

The methodology was agreed upon through an inception phase that established a shared understanding among Plan International Kenya and the consultant on the objectives, scope and analytical expectations of the PCVA and WAP processes. An inception meeting held on 5th September 2025 brought together the COSME project teams to refine the methodological sequence, expected results, deliverable structure and timelines. During this phase, existing climate risk information for Kilifi County was reviewed, data gaps specific to Mwarakaya Ward were identified, and a sequenced work plan was developed outlining participatory diagnostics, household surveys, spatial and climate analysis, and subsequent synthesis and validation activities.

Following institutional alignment, data collection instruments were developed and technically refined to ensure that they were methodologically sound, gender-responsive and culturally appropriate. The instruments included ward-level FGD guides segregated for women, men and youth; participatory mapping and historical timeline guides; seasonal calendar and daily clock templates; household decision-making pile-sorting tools; Leaky Bucket diagrams; impact chain protocols; vulnerability matrix templates; institutional Venn diagram formats; and adaptation pathway worksheets. Each tool integrated prompts on gender norms, intersectional vulnerability, unpaid care workloads, labour shifts during climate stress and disability inclusion.

Stakeholder mapping during the inception phase identified relevant actors within Mwarakaya Ward, including the Ward Climate Change Planning Committee, chiefs and assistant chiefs, village elders, leaders of women and youth groups, PAD representatives and support networks, and community-based organizations. These actors facilitated community mobilization, venue preparation and dissemination of study information at ward level. The inception phase ensured that ward stakeholders were adequately oriented and positioned to meaningfully participate in the PCVA and adaptation planning process.

1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA)

The Ward Adaptation Plan was informed by a comprehensive PCVA process implemented in four COSME target villages within Mwarakaya Ward i.e., Mbuyuni, Ondoni, Silala and Vwevwesi. The assessment followed a structured six-step participatory sequence designed to diagnose climate vulnerabilities, identify underlying drivers of sensitivity, assess existing coping and adaptation strategies, and translate diagnostic findings into actionable adaptation options suitable for inclusion in the WAP. The process applied the full suite of ten PCVA tools, each deployed to generate gender- and age-disaggregated evidence capable of capturing complex interactions between climate hazards, livelihoods, natural resources and social systems.

1.2.2.1 Step 1: Scoping

During scoping, the geographical scope of the ward was delineated using GIS shapefiles provided by Plan International and Kilifi County Government. Within this boundary, the four COSME villages were purposively selected on the basis of ecological sensitivity, concentration of climate-sensitive livelihoods, recurrent exposure to droughts and floods, susceptibility to Pests and diseases outbreaks, and documented human–wildlife conflict incidents, were then developed to summarize demographic characteristics, livelihood systems, existing infrastructure, ecosystem conditions, climate-related impacts, past interventions and vulnerability hotspots. These profiles served as the baseline data against which PCVA tools were deployed.

1.2.2.2 Step 2: Establishment of Core Participatory Analysis Teams

A ward-level core participatory analysis team was established for Mwarakaya, drawing membership from Ward Climate Change Planning Committee (WCCPC) representatives, village elders, women leaders, youth leaders, PAD representatives and carers, NbS groups, GJJ groups, and other active CBOs and self-help groups. County technical officers from climate change, environment, forestry, agriculture, livestock and fisheries were engaged as resource persons.

The core team underwent a structured orientation on the PCVA framework, the ten PCVA tools, gender- and age-responsive facilitation, child safeguarding and ethical data collection. Training sessions used a combination of plenary presentations, group work and simulations of each PCVA tool, conducted primarily in Kiswahili to ensure comprehension. By the end of the training, the ward team had the capacity to co-facilitate hazard and resource mapping, historical timelines, seasonal calendars, daily clocks, household decision making and pile-sorting exercises, Leaky Bucket analyses, impact chain mapping, vulnerability matrix scoring, Venn diagrams and adaptation pathway development.

1.2.2.3 Step 3: Situation Analysis of Climate Vulnerability

The situation analysis integrated secondary datasets, primary field evidence and participatory analytics to generate a comprehensive picture of exposure, sensitivity and adaptive capacity across Mwarakaya Ward. This phase synthesised literature review, ward-level gender- and age-disaggregated Focus Group Discussions (FGDs), transect walks across the four COSME villages, and the structured application of PCVA Tools 1–7. Through triangulation, these methods produced a detailed, spatially nuanced and socially differentiated understanding of how climatic hazards interact with biophysical systems, livelihood structures and gendered social dynamics.

1.2.2.3.1 Literature Review

The literature review served as the technical backbone of the situation analysis, drawing from global, regional, national, county and historical ward-level evidence to contextualize observed vulnerabilities. At the global scale, the review considered findings from IPCC AR6 Working Group I and II, particularly projections relevant to East African coastal systems, including accelerated warming rates, sea surface temperature rise, intensification of short-duration

extreme rainfall, increased occurrence of coastal storms, saline intrusion risks and anticipated sea-level rise along the Western Indian Ocean. These findings were aligned with global adaptation and risk governance frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030, the Paris Agreement adaptation cycle, and SDGs 2, 5, 6, 13 and 14, which collectively frame resilience building in coastal and marine socio-ecological systems.

At the regional scale, the review integrated IGAD Climate Prediction and Applications Centre (ICPAC) bulletins and the Africa Climate Policy Centre (ACPC) projections for Eastern Africa. This regional evidence positioned Mwarakaya within a broader coastal climate risk corridor characterized by rising temperatures, altered monsoon circulation and changing upwelling systems.

National climate policy and regulatory frameworks were examined comprehensively, including the Climate Change Act (2023), National Adaptation Plan (NAP 2015–2030), Kenya's Updated NDC (2023), the National Framework for Climate Services (NFCS), Kenya Climate-Smart Agriculture Strategy, National Disaster Management Policy, National Drought Early Warning System, and EMCA (2015) Cap 387 of the Laws of Kenya and sectoral regulations. These documents provided the lens for interpreting institutional mandates, priority hazards and adaptation responsibilities that cascade to county and ward levels.

At county scale, the review consolidated information from the Kilifi County Integrated Development Plan (CIDP III, 2023–2027), the Kilifi County Climate Change Act (2021), the County Climate Change Action Plan (CCCAP 2023–2027), the County Spatial Plan, and sector strategies in agriculture, livestock, water and coastal management. These county frameworks were triangulated with socio-economic and food security trends reported by the National Drought Management Authority (NDMA), historical rainfall data from the Kenya Meteorological Department (KMD), forestry and mangrove assessments by the Kenya Forest Service (KFS), and demographic data from the Kenya Population and Housing Census (2019).

The review also incorporated previous PCVAs conducted under COSME and other coastal adaptation programmes, which provided comparative vulnerability patterns, gender-responsive insights, and past lessons on nature-based solutions, community-based early warning systems, and local adaptation practices. This multilevel synthesis created a scientifically grounded baseline against which primary data from Mwarakaya could be interpreted.

1.2.2.3.2 Focus Group Discussions (FGDs)

Ward-level FGDs were conducted with segregated groups of women, men and youth from the four COSME villages; Mbuyuni, Ondoni, Silala and Vwevwesi, to elicit gendered and generational insights into climate trends, livelihood impacts and socio-ecological vulnerability (Figure 3). The discussions explored perceived changes in rainfall reliability, including delayed onset and inconsistent cessation of rains, prolonged intra-seasonal dry spells, unprecedented periods of elevated temperatures and shifts in wind patterns. Participants described the material impacts of these climatic variations on agricultural productivity, Kaya forests health, livestock mortality and food security.

The FGDs also highlighted the deeply gendered nature of climate vulnerability. Women described increased walking distances and time burdens for water and fuel collection during droughts; heightened caregiving demands during disease outbreaks; increased exposure to GBV during water scarcity periods; and disruptions in women's informal trading activities. Youth reported loss of wage labour opportunities during climate shocks, stress-related migration, and lack of access to climate-resilient livelihood assets. Men described reduced crops and livestock yields and increased anxiety over household provisioning responsibilities. In addition,

participants highlighted PAD-specific vulnerabilities, including mobility limitations during floods and exclusion from labour-intensive coping strategies.

These discussions produced climate calendars, gender-disaggregated vulnerability matrices and hazard trend narratives that deepened the temporal and social dimensionality of the vulnerability assessment.

Figure 3: Focus Group Discussions in Mwarakaya Ward (Source: Field Surveys,2025)

1.2.2.3.3 Transect Walks

Transect walks undertaken across the four villages provided on-ground spatial verification of vulnerabilities identified in FGDs and documented in literature. These walks traversed rainfed agricultural plots, irrigated horticultural fields, Kaya forests, boreholes and shallow wells, public institutions (schools, health dispensaries), access roads, market areas, and known flood plains.

Structured observation sheets and GPS-enabled devices were used to capture geospatial attributes of key vulnerability hotspots. Evidence recorded included deterioration of feeder roads due to repeated flooding, heavily eroded gullies formed by runoff from bare catchments, degraded Kaya Forest areas exhibiting canopy thinning, and infrastructure damage such as washed-out culverts and submerged pathways.

These observations enhanced the physical realism of the vulnerability analysis and provided geo-referenced evidence that later strengthened hazard–resource maps and the identification of priority zones for ecological restoration and climate-resilient infrastructure.

1.2.2.3.4 Tool 1: Hazard and Resource Mapping

The hazard and resource mapping tool enabled community members to visually represent their environmental and socio-economic landscape (Figure 4). Through participatory sketching, communities identified settlement clusters, farmlands, grazing areas, rivers, markets, schools, health facilities, water points and religious centres. Hazard features; such as recurrent droughts, flood zones, pests and diseases outbreaks and human-wildlife conflicts were marked and discussed. This mapping exercise generated spatially explicit knowledge that later informed adaptation targeting and zoning.

Figure 4: Hazard and Resource Mapping in Ondoni Village, Mwarakaya Ward (Source: PCVA, 2025)

1.2.2.3.5 Tool 2: Historical Timelines

Historical timelines reconstructed multi-decadal climatic and socio-economic events including major droughts, destructive floods, pest invasions (armyworms, whiteflies) and episodes of heightened human–wildlife conflict. Each event was discussed in terms of magnitude, social and economic impact, community responses and institutional support. Timelines revealed a pattern of intensifying climatic shocks, shortened recovery cycles and cumulative stress on natural resource systems.

1.2.2.3.6 Tool 3: Seasonal Calendars

Seasonal calendars detailed intra-annual cycles of rainfall, temperature, wind regimes, tidal patterns, agricultural operations including land preparation, planting, weeding and harvesting, disease outbreaks and social events. Comparisons between historical and current seasonal behaviour revealed declining predictability of rainfall, increased occurrence of off-season rainfall, persistent dry spells and shifts farming seasons. Calendars also captured periods of heightened GBV, school absenteeism, food scarcity and increased care burdens.

1.2.2.3.7 Tool 4: Daily Clock

The daily clock exercise revealed gendered labour allocation and how climate stress reconfigures time use. Under climate-normal conditions, tasks such as farming, childcare, water collection and marketing followed relatively stable patterns. Under climate stress, women and girls reported significant increases in unpaid care work and water collection time, men noted prolonged periods of income loss due to reduced farm yields, and youth described increased domestic workload substitution and reduced learning hours. These insights demonstrated how climate shocks deepen time-poverty and reduce adaptive capacity.

1.2.2.3.8 Tool 5: Household Decision-Making Pile Sorting

Pile sorting exercises clarified intra-household power/authority structures under normal and climate-stress scenarios. Under normal conditions, decision-making on crop choices, livestock sales and minor household investments tended to be joint or gender-specific depending on cultural norms. Under climate stress, however, the decision space often shifted toward male dominance, particularly in decisions involving financial risk, asset disposal or climate-response investment, which in turn constrained women's adaptive agency and reinforced gender-based vulnerabilities.

1.2.2.3.9 Tool 6: Leaky Bucket

The Leaky Bucket tool illustrated how climate shocks disrupt household financial stability. Communities mapped income streams such as farming, petty trade, remittances and casual labour, alongside expenditure flows including food purchases, health costs, water, school fees, debt servicing and transport. Climate shocks were shown to reduce inflows drastically while simultaneously raising outflows through rising food prices, repair costs and medical expenses. This widening income–expenditure gap demonstrated the structural economic vulnerability of households.

1.2.2.3.10 Tool 7: Impact Chains

Impact chain analysis depicted the cascading consequences (direct and indirect impacts) of priority hazards. For instance, prolonged drought triggered crop losses, water scarcity, livestock mortality and reduced milk intake, which in turn contributed to household food insecurity, malnutrition risk among children, increased labour migration by men, rising care burdens for women, heightened GBV risk and reduced school attendance. Similarly, flooding damaged homes, destroyed crops, and disrupted local markets, leading to sudden income losses, increased household expenditures, and heightened food insecurity. This tool exposed multi-dimensional vulnerability pathways and supported the design of integrated adaptation strategies.

1.2.2.4 Step 4: Analysis of Causes of Vulnerability (Tool 8 – Vulnerability Matrix)

The analysis of underlying causes of vulnerability was undertaken through the application of the vulnerability matrix (Tool 8), which provided a structured means of examining how different climatic hazards intersect with livelihood assets and socio-ecological systems in Mwarakaya Ward. Hazards such as drought, floods, human–wildlife conflict and agricultural pests and diseases were arranged against critical assets including rain-fed farms, irrigated plots, Kaya Chonyi, livestock, domestic water sources, housing structures, schools, health facilities, feeder roads and market infrastructure. Community members assigned severity ratings for each hazard–asset interaction, supported by qualitative reasoning and lived examples.

The cumulative scoring process produced a ranking that demonstrated, for example, the high susceptibility of mangrove areas and seaweed farms to combined effects of temperature stress, strong winds and changing salinity regimes; the acute vulnerability of rain-fed agriculture to prolonged drought and unpredictable rainfall; and the differentiated impacts of floods on schools and feeder roads which disrupt mobility, market access and educational continuity. This step bridged descriptive hazard analysis with deeper socio-economic and institutional determinants of vulnerability, forming the foundation for identifying priority adaptation pathways

1.3 Linkages to other planning processes

At both the National, County and Ward Levels, several other planning processes exist to which the WAP is linked. These include the National Adaptation Plan 2015-2030 and The Kilifi County Climate Change Act, 2021. The NAP is Kenya's first plan on adaptation and demonstrates the commitment to operationalize the National Climate Change Action Plan by mainstreaming adaptation across all sectors in the national planning, budgeting and implementation processes. The mainstreaming approach recognizes that climate change is a cross-cutting sustainable development issues with economic, social and environmental impacts. Similarly, the CAP aligns with the NAP in terms of content and reporting focusing on Kilifi County circumstances, climate hazards and adaptation actions which are largely derived from PCVAs for the 35 wards. The WAP adaptation plan is prepared considering both the NAP and CAP and aligning adaptation actions accordingly as informed by the village focused PCVAs.

2 Chapter 2: Coordination of Adaptation Actions

2.1 Introduction

Effective coordination is essential in transforming climate adaptation goals into tangible resilience outcomes. In Mwarakaya Ward, climate risks affect various sectors such as agriculture, water resources, health, ecosystems, and infrastructure. The impact of these risks differs among social groups, with women, youth, and other vulnerable populations often facing the greatest hardships. Implementing the Ward Adaptation Plan therefore demands a governance framework that connects community priorities with county planning structures and national climate policies, while ensuring that local actors have access to finance, technical expertise, and reliable climate information. This chapter outlines that coordination framework and its linkage to Kenya's National Adaptation Plan (NAP) 2015–2030. The coordination mechanism emphasizes both horizontal collaborations, bringing together village-level institutions, community-based organizations, private entities, and technical officers within the ward and vertical integration, ensuring that ward priorities are embedded in county climate change plans and budgets, and aligned with the NAP's strategic objectives. These include climate-smart agriculture, sustainable water resource management, ecosystem restoration, disaster preparedness and early warning systems, and health resilience. The outcome is a harmonized structure where institutional roles are clearly defined, duplication of efforts is minimized, and locally led, gender-responsive adaptation initiatives are effectively implemented, monitored, and sustained over time.

2.2 Current Institutional Arrangements

Coordination presently operates across three interlinked tiers. Kenya's national framework for climate change adaptation and resilience is anchored in the Climate Change Act (2023), which provides the legal foundation for coordinated climate action across all levels of governance. The National Climate Change Council and the Climate Change Directorate provide policy direction, coordination, and technical backstopping to ensure that adaptation measures are integrated into national and county development processes. Operational guidance is delivered through the Kenya National Adaptation Plan (NAP) 2015–2030 and the National Climate Change Action Plan (NCCAP) 2023–2027, which outlines priority actions to enhance adaptive capacity and reduce vulnerability across key sectors. Specialized agencies reinforce this institutional architecture: the Kenya Meteorological Department provides forecasts and impact-based early warnings; the National Drought Management Authority (NDMA) leads drought monitoring and response; the Coast Development Authority (CDA) coordinates integrated coastal zone development, conservation, management and sustainable use of coastal and marine resources; and the National Environment Management Authority (NEMA) enforces environmental safeguards for adaptation investments. These agencies coordinate with county structures and engage directly with wards, ensuring that national policies are translated into locally relevant, gender-responsive, and climate-resilient actions.

At county level, Kilifi County Government acts as the bridge between national strategy and local action. The Kilifi County Climate Change Act (2021) provides a legal framework for coordinated climate action within the county, mandating the County Government to establish institutional structures, mobilize and allocate resources for climate adaptation and mitigation, and ensure that vulnerable groups benefit equitably from climate financing. The Act's implementation is operationalized through the Kilifi County Climate Change Action Plan (2023–2027), which outlines strategic interventions aligned with national and county priorities to enhance resilience, promote sustainable livelihoods, and guide sectoral climate responses. Additionally, the County Integrated Development Plan (CIDP) 2023–2027, integrates climate change considerations into broader development planning, ensuring that climate actions are mainstreamed into the county's long-term socio-economic agenda and across various departments (Environment and Climate Change; Infrastructure and Transport; Agriculture and

Blue Economy; Water and Sanitation; Health and Tourism) to integrate resilience criteria into annual work plans. Together, these frameworks position the county in climate governance, fostering inclusive, well-coordinated, and sustainable climate responses. County coordination platforms such as the County Climate Change Steering and Planning Committees oversee the integration of climate adaptation and mitigation across sectors in Kilifi. These structures, supported by the Directorate of Climate Change, coordinate multi-stakeholder engagement, resource mobilization, and the implementation of the County Climate Change Action Plan. Moreover, partnerships with non-governmental organizations, development partners, and private sector actors are convened through county-led forums, promoting alignment of external climate initiatives with county priorities and public investments, thereby enhancing coordination, transparency, and impact.

At the ward and community level, implementation is driven by locally led structures that link planning, decision-making, and on-the-ground action. The Ward Administrator acts as the administrative liaison to county directorates, while the Ward Climate Change Planning Committee (WCCPC) functions as the main coordination body for Mwarakaya Ward. The WCCPC includes local administration, technical extension officers, Water Resource User Associations, Community-Based Organizations, and representatives of women, youth, PADs and respected elders. The committee guides investment prioritization, monitors progress, and facilitates community engagement in adaptation actions. Village-level structures, such as water user committees, disaster volunteer teams, and environmental stewardship groups, execute day-to-day interventions, including maintaining boreholes and managing forest restoration projects. By formalizing these linkages, the WCCPC strengthens communication, ensures accountability, and promotes inclusive decision-making, enabling Mwarakaya Ward to implement adaptation measures effectively while fostering community ownership.

2.3 Relevant Policies

The coordination framework for climate adaptation in Mwarakaya Ward is supported by a strong policy environment. At the national level, the Climate Change Act (2023) and National Climate Change Action Plan (NCCAP 2023–2027) provide the overarching structure for climate governance, emphasizing locally led adaptation and devolved financing. The National Adaptation Plan (NAP 2015–2030) identifies priority sectors relevant to Mwarakaya, including water resource management, climate-smart agriculture, ecosystem restoration, disaster preparedness, and human health resilience. Programs such as the Financing Locally Led Climate Action (FLLoCA) channel grants to support county and ward-level initiatives.

At the county level, the Kilifi County Climate Change Act, 2021 and Kilifi County Climate Change Action Plan, 2023–2027 provide the framework for mainstreaming adaptation into sectoral plans, including water, agriculture, environment, lands, and disaster management. Sector-specific policies further reinforce this framework: the Water Act, 2016 and National Water Policy, 2019 embed community-based water governance and integrated water resource management; the Agricultural Sector Transformation and Growth Strategy, 2019 promotes diversified, climate-resilient production and value addition; the National Disaster Risk Management Policy, 2021 strengthens anticipatory action and local preparedness; the Wildlife Conservation and Management Act, 2013 provides mechanisms for mitigating human–wildlife conflicts; and national gender and social inclusion policies ensure equitable participation and benefit sharing in climate initiatives

2.4 Proposed Coordination Arrangements

The proposed coordination framework for Mwarakaya Ward strengthens existing systems while introducing clear linkages, communication channels, and accountability mechanisms to support coherent and inclusive implementation of adaptation actions. The arrangement mirrors national

expectations for devolved climate governance under the Climate Change Act (2023) and the Kilifi County Climate Change Act (2021), ensuring alignment between community priorities and county-level planning, financing, and reporting processes.

At the ward-county interface, a designated County Climate Change Officer from the County Climate Change Unit (CCCU) will serve as the technical liaison between the WCCPC and the county departments. The officer will attend WCCPC meetings, share updates on ongoing county programs, and support integration of ward priorities into sectoral plans and the County Climate Change Fund (CCCF) proposal pipeline. The WCCPC, in collaboration with the Ward Administrator, will prepare quarterly progress reports summarizing implementation progress, budget utilization, gender-disaggregated data, challenges, and lessons learned. These reports will be submitted to the CCCU to ensure upward accountability and inform county planning and resource allocation.

Within the ward, the WCCPC will act as the central coordination and decision-making platform. It will convene a biannual Mwarakaya Climate Stakeholder Forum, bringing together all relevant actors including county departments, national agencies, NGOs, CFAs, WRUAs, self-help groups, private entities, and academic and research institutions. The forum will serve as a space to map ongoing interventions, harmonize technical standards, identify overlaps or gaps, and develop joint action plans. It will also facilitate knowledge sharing, mobilize resources, and enhance coordination between local and county-level actors.

To strengthen operational efficiency, the WCCPC will establish task-specific subcommittees responsible for thematic areas such as water management, forest and ecosystem restoration, climate-smart agriculture, and disaster preparedness. Each subcommittee will include representatives from relevant community groups, technical officers, and development partners, ensuring that actions are context-specific, technically sound, and gender-responsive. For instance, conservation partners such as WWF-Kenya and Botanical Garden Conservation International (BGCI) to rehabilitate degraded forest areas, promote tree planting, and strengthen community forest governance structures.

Given Mwarakaya's vulnerability to recurrent droughts, pests, and human-wildlife conflicts, climate coordination will be closely linked with disaster risk management structures. The WCCPC will collaborate with the Ward Disaster Management Committee to ensure early warning systems, community awareness, and anticipatory actions are integrated into regular adaptation planning. Seasonal forecasts from the Kenya Meteorological Department (KMD) and early warning information from the NDMA will be disseminated through community barazas, local FM stations, and mobile platforms to enhance preparedness and response.

Finally, an annual review and reflection meeting, co-convened by the CCCU and the WCCPC, will assess progress against the NAP indicators and the County Climate Change Action Plan (2023–2027) targets. This platform will document best practices, challenges, and emerging lessons to improve coordination, refine institutional arrangements, and strengthen adaptive management at the ward level.

2.5 Implementation Roles

Effective implementation of the Ward Adaptation Plan requires clear role delineation among all stakeholders to ensure that adaptation measures are executed efficiently, inclusively, and in line with Kenya's National Adaptation Plan (2015–2030). Table 1 below outlines the mandates, decision-making authority, and accountability mechanisms of the key institutions and actors engaged in rolling out adaptation interventions in Mwarakaya Ward. The WAP explicitly mainstreams Gender Equality and Inclusion (GEI) across all stages of planning and

implementation and establishes strong coordination linkages with health, gender, and protection actors to guarantee continuity of essential Sexual and Reproductive Health and Rights (SRHR) and GBV prevention and response services during climate shocks.

Table 1: Institutional Roles in Implementation of Adaptation Actions

Institution/Actor	Key Roles and Responsibilities
Ward Climate Change Planning Committee (WCCPC)	Lead planning, coordination, and prioritization of ward-level adaptation actions; mobilize community participation; ensure gender and social inclusion in design and implementation
Kilifi County Government Departments	Provide technical guidance, financing, and sectoral oversight (Water, Agriculture, Livestock, Environment, Forestry, Roads, Health, and Education); integrate ward priorities into departmental budgets and work plans
National Drought Management Authority (NDMA)	Coordinate drought early warning and response systems; activate anticipatory action and social protection linkages
Kenya Meteorological Department (KMD)	Provide localized weather forecasts, early warnings, and seasonal outlooks to support agricultural planning and disaster preparedness
Kenya Forest Service (KFS)	Support reforestation, forest management, and training on sustainable forest practices; collaborate with CFAs on forest restoration
National Environment Management Authority (NEMA)	Review and approve Environmental Impact Assessments (EIAs) for adaptation projects; ensure adherence to environmental safeguards.
Development Partners and NGOs (Plan International, WWF-Kenya, BGCI, KCEP, CARE, ActionAid)	Support community-based adaptation initiatives; provide funding, training, and technical assistance in agriculture, water, and ecosystem management
Community-Based Organizations (CBOs), CFAs, and Self-Help Groups	Implement localized projects such as forest restoration, water harvesting, and livelihood diversification; mobilize local labor and knowledge
Private Sector and Cooperatives	Invest in renewable energy, irrigation, and climate-smart technologies; provide financial products and services for resilience building
Academic and Research Institutions (Pwani, KMFRI, KEMFRI, Universities)	Conduct participatory research, monitor ecological change, and support innovation and knowledge sharing on adaptation practices

3 Chapter 3: Situation analysis of climate vulnerability

3.1 Introduction

The ward is increasingly experiencing multiple, interlinked, and climate-induced hazards whose intensity, duration, and frequency have grown over time. Findings from the Participatory Climate Vulnerability and Capacity Assessment (PCVA, 2025) indicate that the community is recurrently affected by drought, floods, pests and diseases, and emerging human–wildlife conflicts (HWCs). These hazards have progressively disrupted ecosystem balance, altered natural resource functions, and reshaped traditional livelihood systems across Ondoni, Silala Mbuyuni and Vwevwesi areas. The assessment employed a combination of community-based knowledge, historical trend analysis, and scientific projections to establish the frequency, intensity, and spatial extent of each hazard. Vulnerability was analyzed across the four sampled villages i.e., Mbuyuni, Ondoni, Silala and Vwevwesi, based on two key parameters: exposure to climatic stressors and adaptive capacity at household and institutional levels.

The results reveal that women, widows, elderly persons, persons abled differently (PAD), and low-income households bear the greatest burden of climate-related impacts. Their heightened vulnerability stems from dependence on climate-sensitive natural resources, limited access to financial and productive assets, and increased unpaid labour associated with caregiving and resource collection during crisis periods. Spatial analysis further shows that Drought predominantly affects villages such as Chengoni, Chije, Mbuyuni, Kizingo, Mitulani, Bandarasalama, and Mwarakaya, indicating localized vulnerability to dry conditions. Pests and diseases are widespread and linked with the extensive agricultural practices throughout the entire ward. Flooding is primarily concentrated along the riverine ecosystems of River Mwanzai and Mto Mkuu, where intense rains cause river overflow and inundation, impacting settlements and farmland. Furthermore, human-wildlife conflicts are most recurrent in Ondoni and Vwevwesi villages, where they cause crop damage and pose safety risks to the community. Understanding these spatial and demographic dynamics provides a crucial evidence base for prioritizing adaptation interventions aimed at minimizing exposure, reducing vulnerability, and enhancing the long-term resilience of communities and ecosystems in Mwarakaya Ward.

The results reveal that women, widows, elderly persons, persons with disabilities (PWD), and low-income households bear the greatest burden of climate-related impacts. Their heightened vulnerability arises from reliance on climate-sensitive natural resources, limited access to financial and productive assets, and increased unpaid labor related to caregiving and resource collection during crises. Spatial analysis further shows that drought predominantly affects villages such as Chengoni, Chije, Mbuyuni, Kizingo, Mitulani, Bandarasalama, and Mwarakaya; Pests and diseases are widespread across the entire ward, linked to extensive agricultural practices. Flooding is mainly concentrated along the riverine ecosystems of River Mwanzai and Mto Mkuu, where heavy rains cause overflow and inundation, impacting settlements and farmlands. Human-wildlife conflicts are most frequent in Ondoni and Vwevwesi villages, causing crop damage and posing safety risks. Understanding these spatial and demographic dynamics provides a crucial evidence base for prioritizing adaptation interventions to minimize exposure, reduce vulnerability, and enhance the long-term resilience of communities and ecosystems in Mwarakaya Ward

3.2 Current climate trends

Mwarakaya Ward experiences a tropical coastal climate characterized by bimodal rainfall and consistently warm temperatures throughout the year. The current climatic conditions in Mwarakaya Ward, based on the baseline analysis under the SSP2–4.5 scenario, indicate a moderately wet environment with annual rainfall averaging 1,258.7 mm, against a potential evapotranspiration (PET) of 1,405.6 mm and an actual evapotranspiration (AET) of 1,662.4 mm (Figure 5). This climatic balance demonstrates that while Mwarakaya receives a fair amount

of rainfall annually, the higher evapotranspiration rates indicate greater atmospheric water demand than rainfall supply. As a result, the ward periodically experiences soil moisture stress and water scarcity, particularly during dry spells, underscoring the need for improved water conservation and climate-resilient agricultural practices.

Temperature analysis under the same scenario (Figure 6) shows a mean air temperature of 25.4 °C and a land surface temperature (LST) of 35.2 °C, indicating the presence of considerable surface heating across terrestrial landscapes. This warming trend reflects progressive surface heating across terrestrial landscapes, influenced by reduced vegetation cover, land degradation, and expanding human settlements. The combination of elevated temperatures and moderate rainfall suggests heightened evapotranspiration rates, which may increase crop-water requirements and reduce the efficiency of rain-fed agriculture

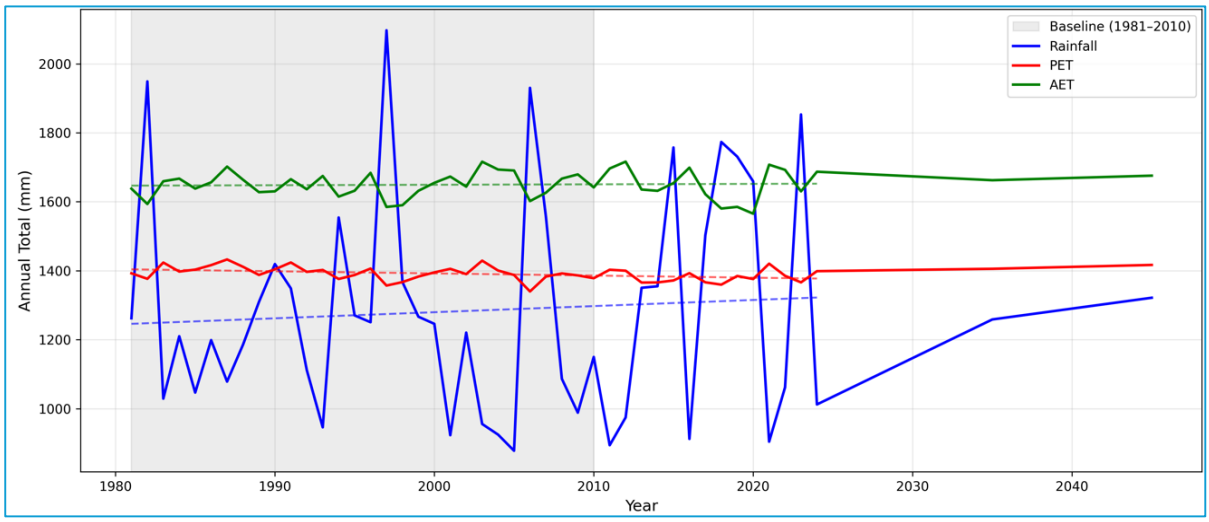

Figure 5: SSP2 4.5 Rainfall, AET and PET Projections (Source: Chirps database and ERA5-Ag, 2022)

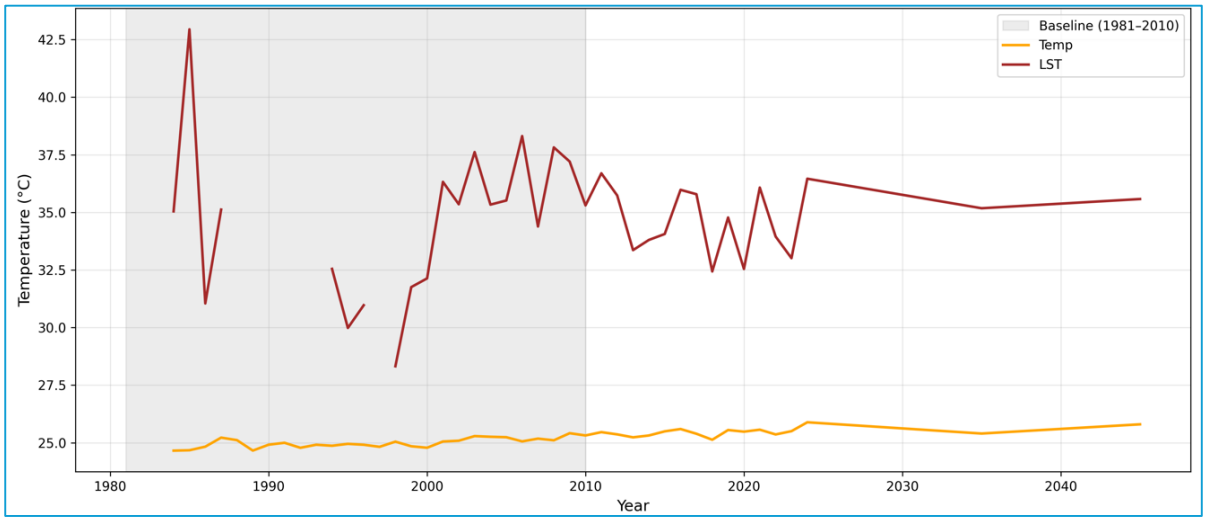

Figure 6: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.3 Future climate trends

Projected climate conditions under the high-emission scenario (SSP5–8.5) indicate that Mwarakaya Ward will experience slightly wetter and warmer conditions relative to the baseline period. Rainfall is projected to increase from 1,321.6 mm to 1,384.5 mm, potential evapotranspiration (PET) from 1,416.7 mm to 1,436.3 mm, and actual evapotranspiration (AET)

from 1,675.6 mm to 1,698.7 mm (Figure 7). While total precipitation shows a modest upward trend, the consistently higher PET values compared to rainfall indicate that atmospheric water demand will continue to surpass precipitation input. This suggests that, despite slightly improved rainfall, hydrological stress and soil moisture deficits will persist, particularly during dry seasons.

Future temperature forecasts underscore continued warming. The mean air temperature is expected to rise from 25.8 °C to 26.0 °C, while the LST increases significantly from 35.6 °C to 35.8 °C (Figure 8). This increase in surface and air temperatures is attributed to intensified greenhouse gas emissions, declining vegetation cover, and expanding land-use changes. The resultant warming is likely to elevate evapotranspiration rates, intensify crop-water demand, and heighten heat stress for both humans and livestock.

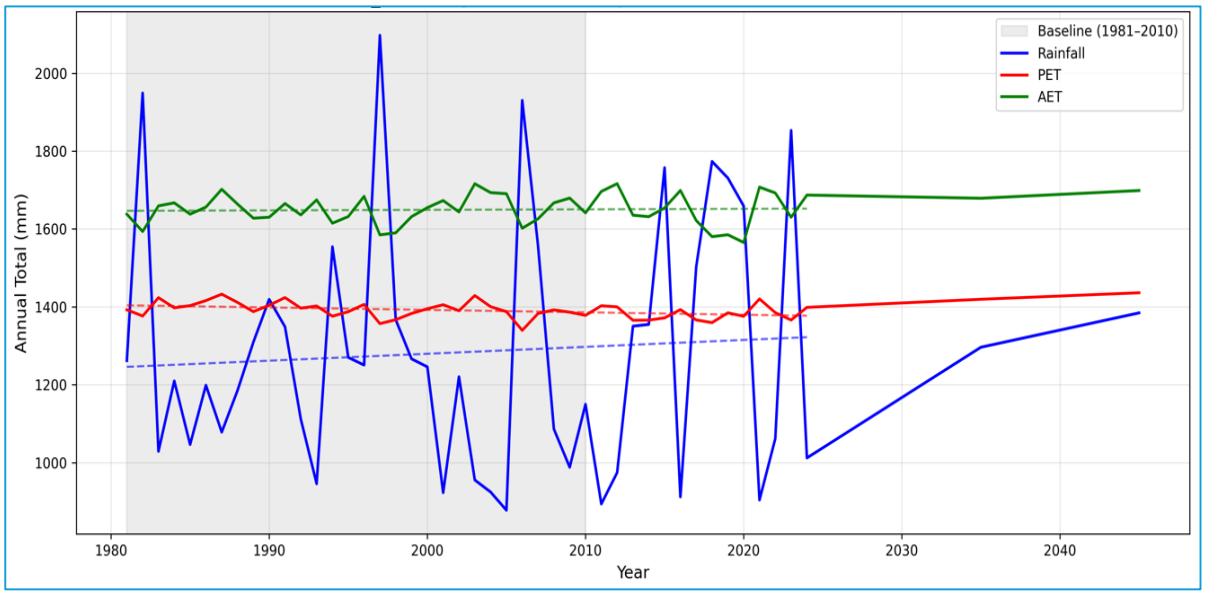

Figure 7: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022)

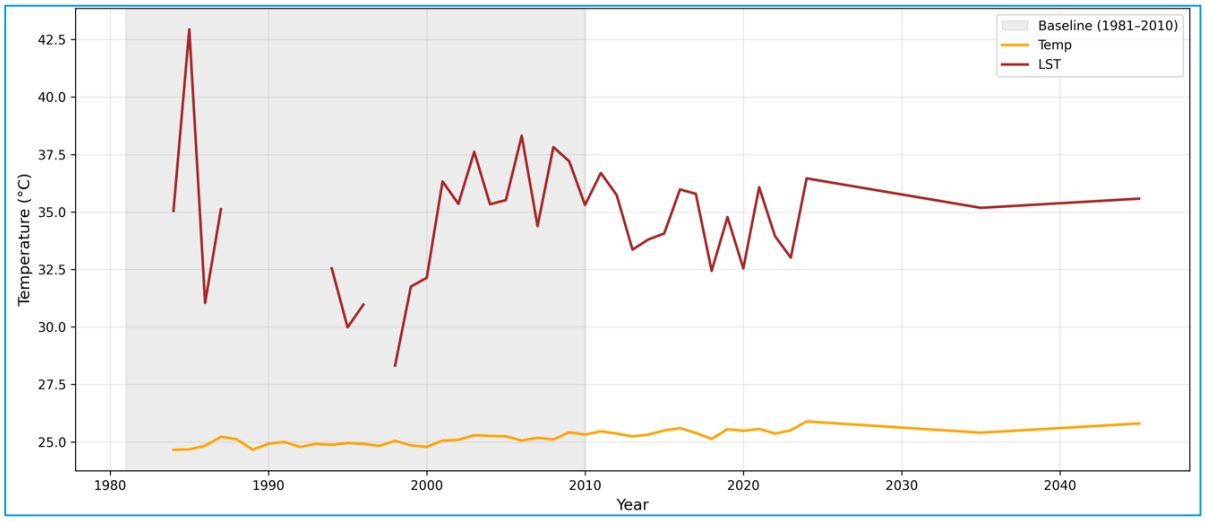

Figure 8: SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.4 Climate hazards and vulnerability

3.4.1 Drought

Drought is the most recurrent climatic hazard in Mwarakaya Ward, impacting both ecosystems and livelihoods. Historical records from 1930 to 2025 show a progressive increase in drought frequency with major drought events occurring at least once every three years. Severe events were recorded in 1930, 1948, 1960, 1978–1982, 2017, and 2024 (Figure 9). Over time, there has been an increase in the frequency, duration, and severity of these droughts, reflecting decreasing rainfall reliability and prolonged dry spells between seasons. This pattern is linked to delayed onset of rains, earlier end of the long rains, and increased evapotranspiration rates, all contributing to shorter growing seasons and frequent crop failures.

Spatial vulnerability is not uniform across the ward. Vwevwesi village, with a vulnerability score of 39, exhibits the highest exposure levels due to its dependence on rain fed agriculture and limited alternative sources of livelihoods. Ondoni and Mbuyuni villages, with moderate vulnerability scores of 27 and 25 respectively, face moderate exposure, indicating some level of resilience possibly supported by diversified livelihoods or better access to resources. Silala village, with the lowest score of 18, experiences comparatively lower vulnerability, likely benefiting from either more stable water sources (Figure 10).

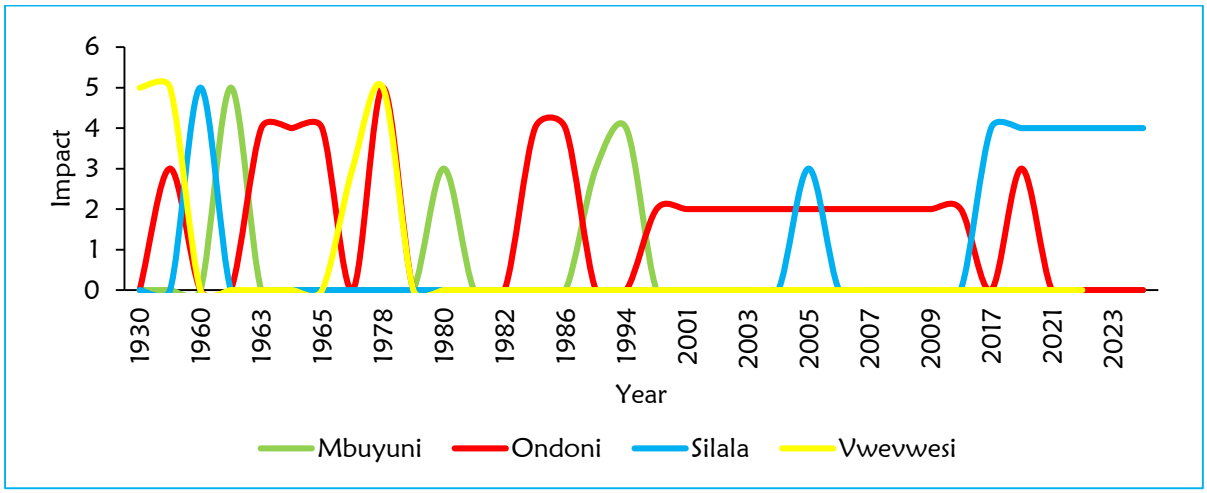

Figure 9: Drought prevalence historical trend (Source: PCVA, 2025)

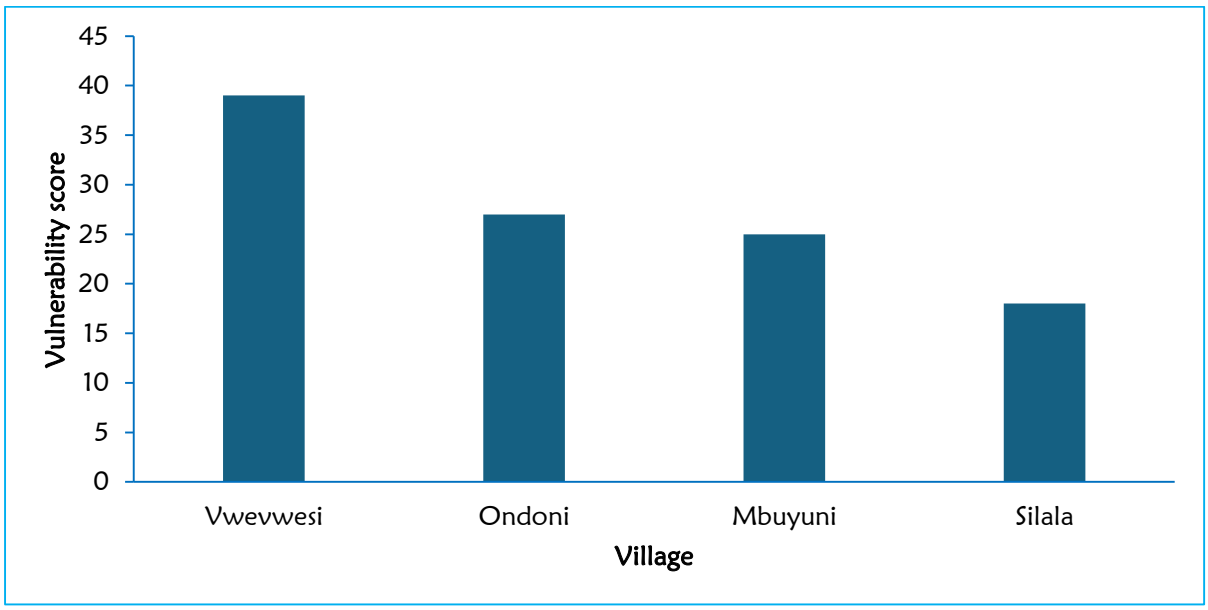

Figure 10: Drought vulnerability score in Mwarakaya ward (Source: PCVA, 2025)

Households situated on higher, less vegetated grounds experience more severe water scarcity, whereas communities near low-lying mangrove and estuarine zones are indirectly affected through declining water tables and saline intrusion into wells. Prolonged drought has led to the degradation of Kaya Chonyi, resulting in the loss of indigenous tree species, decreased forest cover, and disruption of natural water source recharge. The NDVI map of Kaya Chonyi shows a clear seasonal difference in vegetation health and greenness between the dry and wet seasons. During the dry season, NDVI values range from 0.30 to 0.49, indicating reduced vegetation vigor due to limited soil moisture and high evapotranspiration stress, typical of prolonged dry spells. In contrast, the wet season NDVI values increase to 0.24–0.60, reflecting enhanced vegetation productivity and canopy density resulting from increased rainfall and soil moisture availability (Figure 11). This seasonal variation illustrates the sensitivity of vegetation in Kaya Chonyi to changing climatic conditions. Increasingly erratic rainfall patterns and prolonged droughts may lower overall NDVI values over time, signaling declining vegetation resilience and gradual land degradation in the area.

At the community level, drought impacts are strongly gendered and socially differentiated. Women and youth are disproportionately burdened as they bear the primary responsibility for water collection, fuelwood gathering, and household food provisioning. During prolonged drought periods, women are forced to travel for longer distances daily in search of portable water, reducing their time for income-generating activities and increasing exposure to heat-related health risks. Men, on the other hand, experience livelihood disruptions due to reduced agricultural productivity and loss of income from the sale of firewood. Elderly persons and persons abled differently (PAD) face heightened vulnerability as they have limited mobility and adaptive capacity.

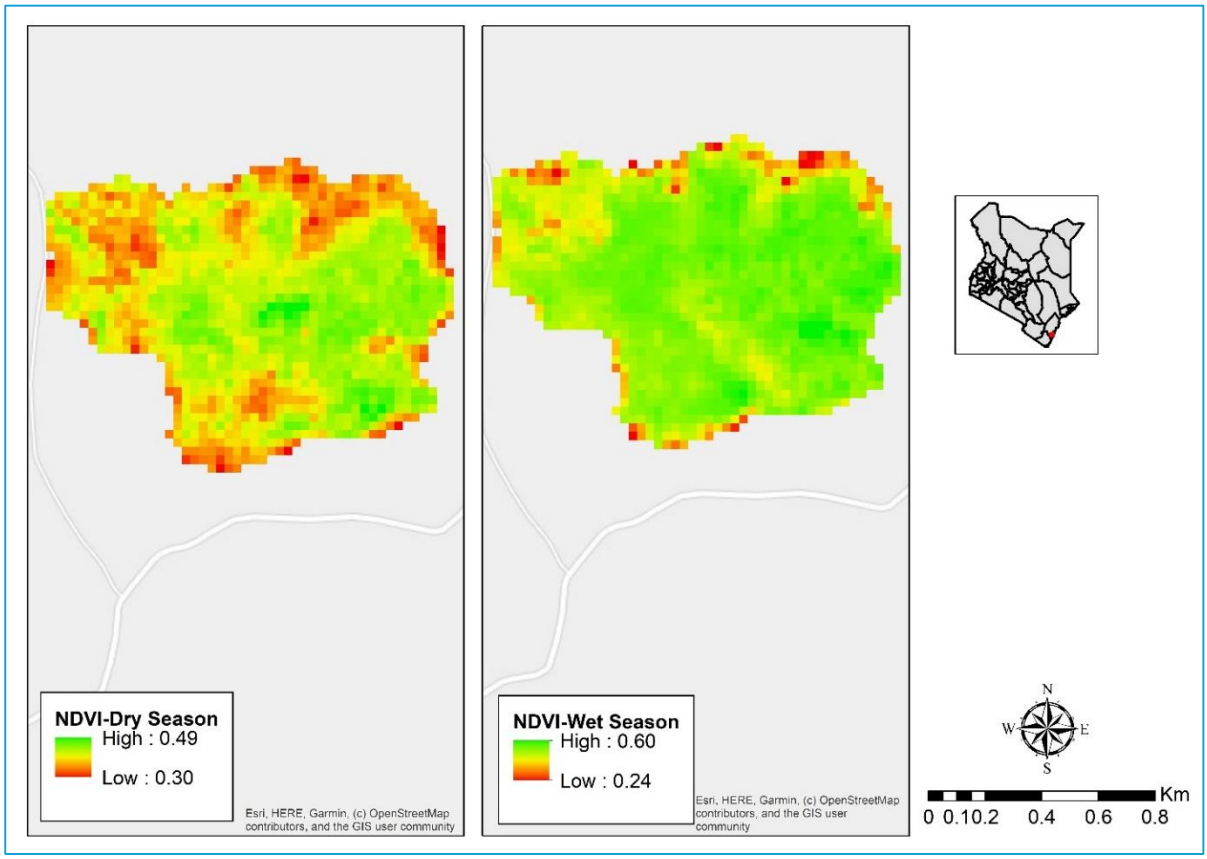

Figure 11: Kaya Chonyi NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2024)

3.4.1.1 Exposure to drought

Exposure to drought in Mwarakaya Ward is extensive and cross-cutting, affecting both ecological and livelihood systems across all four villages. The most prone drought hotspots include Chengoni, Chije, Mbuyuni, Kizingo, Mitulani, Bandarasalama, and Mwarakaya villages indicating localized vulnerability to dry conditions. Water resources such as water pans, boreholes and shallow wells dry out entirely or yield less water due to lowered groundwater levels while River Mwanzai and Mto Mkuu reduce significantly in flow, disrupting ecosystems and the availability of surface water for communities and wildlife. Decreased water availability intensifies competition among households resulting in conflicts over the limited water sources. Consequently, Agricultural productivity decreases intensifying poverty. Kaya Chonyi is highly exposed due to the drying of understory vegetation which accelerates soil erosion and loss of forest biodiversity. Livestock exposure is particularly severe in areas where prolonged scarcity of pasture and water leads to increased mortality rates, reduced milk production, and recurrent livelihood losses. This wide-ranging exposure underscores the interconnectedness of drought impacts on both natural and human systems in Mwarakaya Ward. drought exposure severely limits access to essential water sources like water pans, boreholes, and rivers, causing reduced water availability for domestic use, agriculture, and livestock. This scarcity leads to declining agricultural productivity, livestock mortality, and income losses, heightening poverty and food insecurity. Competition over limited water resources intensify social conflicts among households, while environmental degradation, such as soil erosion and loss of biodiversity in Kaya Chonyi, further undermines community livelihoods and resilience.

3.4.1.2 Socio-Demographic Vulnerability to Drought

Vulnerability to drought in Mwarakaya Ward exhibits marked socio-demographic disparities shaped by gender, age, and livelihood dependence. Women, particularly those heading households, are disproportionately impacted as they shoulder the primary burden of securing water, food, and fuelwood; during prolonged dry spells, they walk longer distances to reach water sources, heightening their physical and economic strain. Youth, especially those engaged in casual labour, experience significant livelihood disruptions as water scarcity and heat stress reduce productivity and income opportunities. Elderly persons and persons abled differently (PAD) remain highly vulnerable due to limited mobility and reliance on others for access to water and food resources. In adaptive terms, only a small fraction of households, mainly male-headed or those with diversified incomes from trade, livestock, or formal employment, are coping effectively with recurrent droughts, while the majority of smallholder farmers are declining, facing shrinking yields and deteriorating living conditions. A few households supported by ongoing interventions, such as water harvesting and mangrove restoration programs, demonstrate improving resilience, adopting practices that stabilize livelihoods despite the increasing drought frequency. This uneven capacity to withstand and recover from drought underscores the need for gender-responsive and socially inclusive adaptation strategies within the ward.

3.4.2 Pests and Diseases

Pests and diseases outbreaks have become an increasingly frequent climate induced hazard across the ward, with clear linkages to changing temperature, humidity and rainfall regimes. Historical analysis from 1978 to 2025 shows that pests and diseases outbreaks occur at least once every two years with periods of high vulnerability observed from 1978 to 1998 ,2013 and 2019 (Figure 12) The frequency and distribution of these outbreaks have intensified over time due to erratic rainfall patterns, warmer temperatures, and prolonged dry spells that weaken crop and livestock resistance. Vulnerability analysis shows that Vwevwesi (27) was most affected, followed by Ondoni (16) and Mbuyuni (13). (Figure 13). These villages have reported widespread infestations including fall armyworm on maize, fruit flies on mangoes, and cassava

mosaic virus, leading to significant crop losses and reduced agricultural yields, while livestock diseases such as foot-and-mouth and tick-borne illnesses have incurred economic losses

Natural ecosystems and key livelihood resources across the ward have been equally vulnerable. Kaya Chonyi which provides essential ecosystem services such as soil stabilization and microclimate regulation has increasingly suffered from pest infestations and fungal infections. These biological threats cause extensive leaf defoliation, weakening the trees and reducing their ability to photosynthesize effectively. As a result, forest regeneration processes are impaired, leading to slower recovery rates and potential long-term degradation of the forest structure threatening biodiversity while undermining the ecosystem services provided. At the community level, exposure to pests and diseases is shaped by socio-economic conditions and gender dynamics, particularly among households engaged in subsistence farming. Farmers who rely on seasonal rainfall for crop production are highly exposed to pest infestations and plant diseases often leading to significant yield losses, directly threatening household food security and income. Women, who play a major role in subsistence farming and post-harvest handling, face disproportionate impacts due to reduced harvests and increased workload required for replanting or pest management. Pastoral households, especially in Mbuyuni, Silala and Vwevwesi, are affected by recurring livestock diseases such as foot-and-mouth and tick-borne infections, which intensify during wetter years following prolonged droughts.

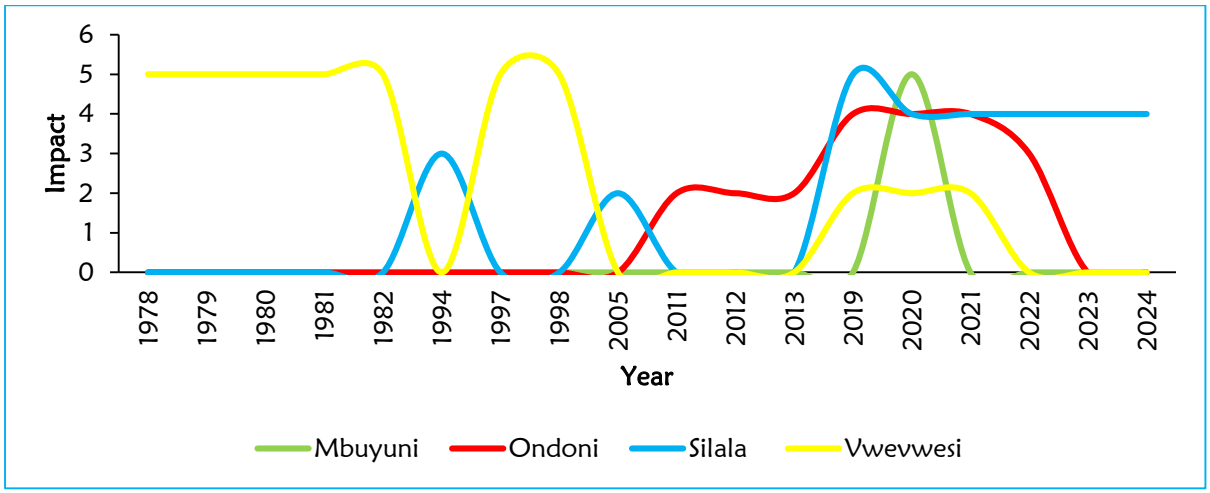

Figure 12: Pests and diseases prevalence historical trend (Source: PCVA, 2025)

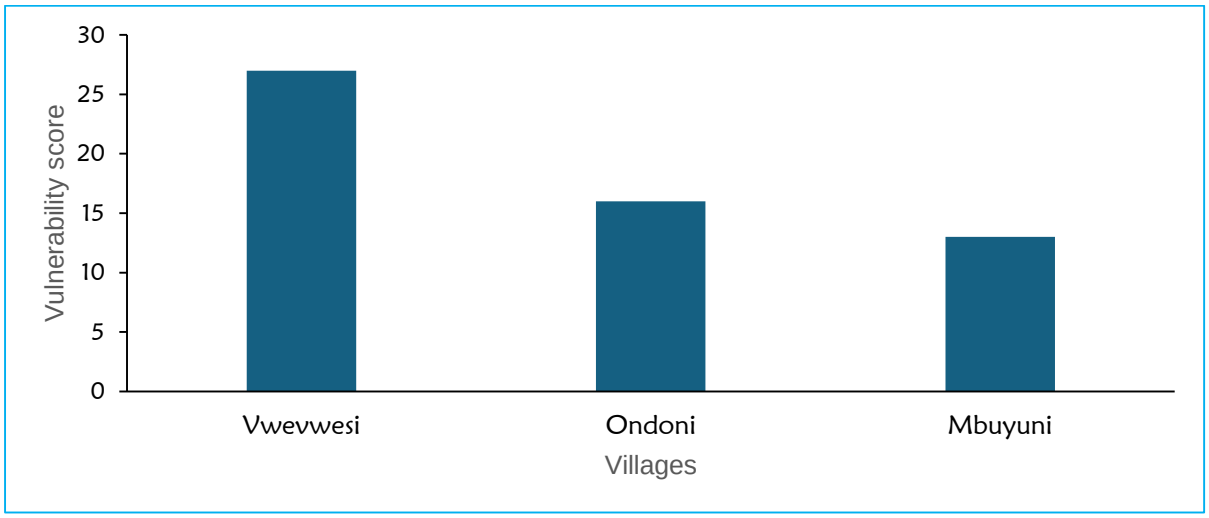

Figure 13: Pests and diseases vulnerability score in Mwarakaya Ward (Source: PCVA, 2025)

3.4.2.1 Exposure to Pests and Diseases

Exposure to pests and diseases in Mwarakaya Ward is widespread, affecting all villages and livelihood systems. The entire ward faces recurrent infestations and outbreaks that impact crops, livestock, and human health due to changing climatic conditions, particularly fluctuations in temperature, rainfall, and humidity. Periodic pest invasions such as fall armyworms and fruit flies have devastated farmlands in Vwevwesi, Ondoni and Mbuyuni while livestock diseases such as foot-and-mouth and tick-borne infections have increased in frequency following alternating drought and wet seasons. This exposure is heightened by the ward's dependence on rain-fed agriculture, open grazing systems, and the proximity to degraded forest and wetland habitats that harbour pest breeding sites.

3.4.2.2 Socio-Demographic Vulnerability to Pests and Diseases

Socio-demographic vulnerability to pests and diseases in Mwarakaya Ward is influenced by factors such as livelihood dependence, gender roles, age, and access to adaptive resources. Smallholder farmers and pastoralists, whose livelihoods depend directly on climate-sensitive natural resources, are the most affected groups. Women, who constitute a significant proportion of these groups, are particularly exposed as they manage subsistence farms and tend small livestock, that are highly susceptible to outbreaks of pests and diseases. Infestations such as fall armyworms and fruit flies, significantly reduce harvests, undermining women's income and household food security. Their capacity to adapt is further limited by restricted access to pest control technologies, extension services, and credit facilities, often leaving them reliant on traditional coping mechanisms that may no longer be adequate against increasingly frequent and severe climate shocks.

Youth face heightened vulnerability due to dependence on casual labour and low-income agriculturally based work, which are directly affected by pest and disease outbreaks. Disruption of these income sources often leads to unemployment and migration pressures. Elderly persons and persons abled differently (PAD) are also highly vulnerable, as physical limitations restrict their participation in pest management activities or relocation during severe outbreaks. In villages such as Mbuyuni, Silala and Vwevwesi, livestock diseases, including foot-and-mouth and tick-borne infections, repeatedly deplete household assets, disproportionately affecting poorer families who lack access to veterinary services or the means to restock animals.

From a health perspective, children and the elderly are particularly susceptible to secondary disease outbreaks, including malaria, dengue, and chikungunya, which often accompany climatic stress and increased pest proliferation. In contrast, a minority of households with diversified income sources or access to agricultural support services demonstrate moderate coping capacity, utilizing improved crop varieties and pesticides. However, the majority of vulnerable groups experience repeated losses that progressively erode resilience. These patterns highlight entrenched gender and socio-economic inequalities and highlight the need for inclusive, gender-responsive adaptation strategies that improve access to climate-smart agricultural practices, strengthen pest and disease monitoring systems, and enhance community-level capacity to cope with recurrent pest and disease pressures in Mwarakaya Ward.

3.4.3 Floods

Historical flood records between 1970 and 2025 indicate major flood occurrences in 1961, 1965, 1972, 1997, 2018, and 2023, translating to an average recurrence interval of approximately one significant flood every 6 years (Figure 14). However, the frequency and magnitude of floods have intensified in recent decades due to increased rainfall variability, catchment degradation, and expansion of settlements into flood-prone zones. Vulnerability analysis denotes Ondoni (32), Silala (25), and Mbuyuni (21) are greatly exposed to flooding with increased damage to farmlands, settlements and infrastructure (Figure 15).

The effects of floods on natural and livelihood resources are widespread and multifaceted. Kaya Chonyi experience soil erosion, excessive surface runoff, and loss of ground cover, undermining ecological functions such as soil stabilization, groundwater recharge, and hydrological regulation. Along the riverine ecosystems of River Mwanzai and Mto Mkuu flooding results in severe land degradation, destruction of adjacent farmlands and settlements as well as the loss of livelihoods

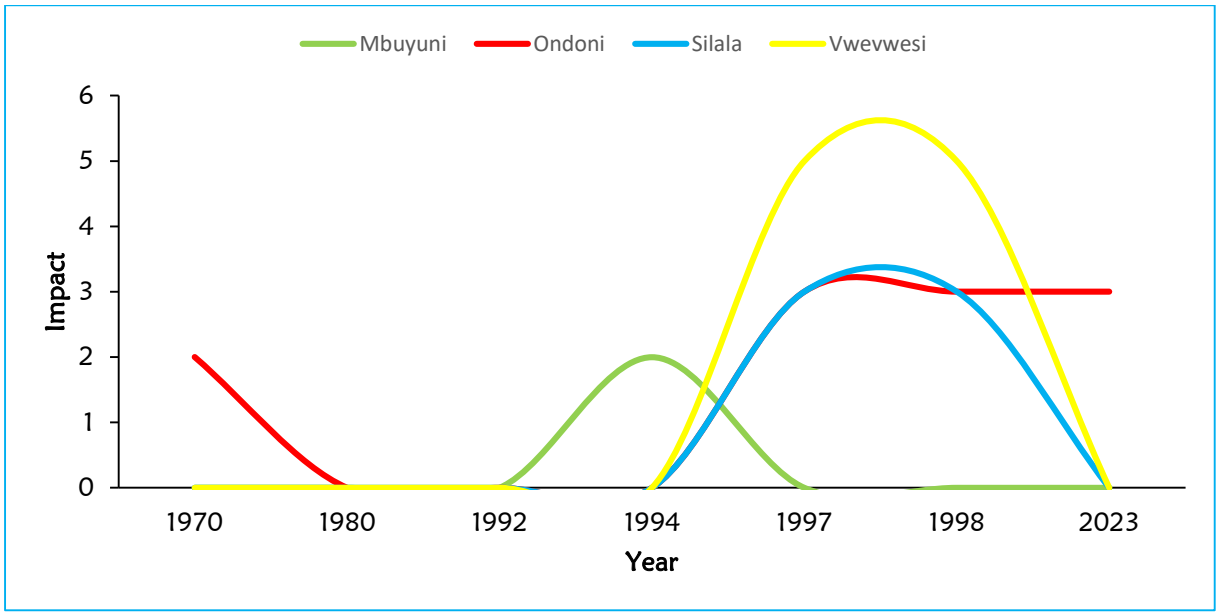

Figure 14: Floods prevalence historical trend (Source: PCVA, 2025)

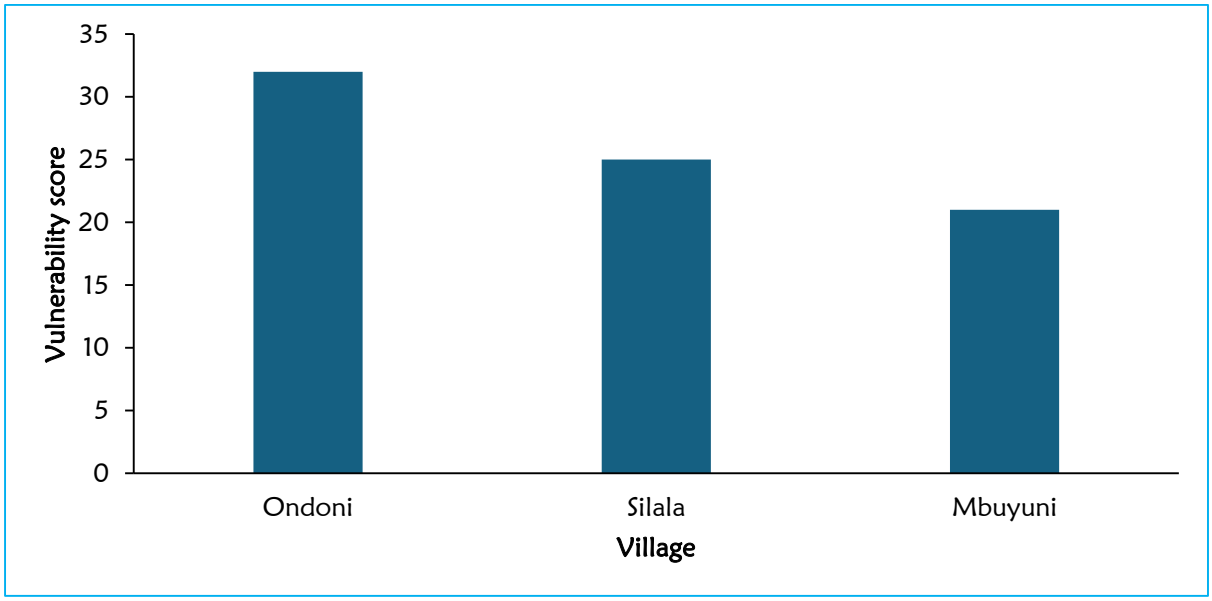

Figure 15: Floods vulnerability score in Mwarakaya Ward (Source: PCVA, 2025)

3.4.3.1 Exposure to Floods

Exposure to floods in Mwarakaya Ward is concentrated along the riverine ecosystem of River Mwanzia and Mto Mkuu with the most flood exposed areas i.e., Ondoni, Mbuyuni and Silala experiencing extensive inundation. This flooding causes destruction of farmlands, loss of crop yields, and erosion of access roads, which disrupts mobility and isolates communities during peak rainfall seasons. Consequently, food production is compromised, intensifying food insecurity as farming households reliant on seasonal cultivation are displaced due to periodic

submersion of their farmlands. The convergence of these flood-prone zones illustrates a complex interplay between altered hydrological patterns, land-use degradation, and unregulated settlement expansion. This combination amplifies the vulnerability of socio-economically marginalized and resource-dependent communities in Mwarakaya Ward, increasing their risk to recurrent flooding hazards and associated socio-economic impacts.

3.4.3.2 Socio-Demographic Vulnerability to Floods

Socio-demographic vulnerability to floods in Mwarakaya Ward is shaped by gender, age, physical ability, livelihood type, and socio-economic status, all of which influence how different groups experience and recover from flood events. Women, particularly those heading households or engaged in informal trade, face heightened vulnerability due to their domestic responsibilities and limited access to productive assets. During flood periods, they bear the additional burden of caring for sick family members, collecting safe water from distant sources after wells become contaminated, and managing food shortages when access to markets and farmlands is disrupted. Children are similarly at high risk owing to exposure to waterborne diseases such as cholera and diarrhoea, unsafe play in flooded environments, and interruption of schooling when learning institutions are used as temporary shelters. Elderly persons and persons abled differently (PAD) face unique challenges related to mobility, evacuation, and access to medical services, making them more dependent on others for survival during emergencies.

Economically, low-income households, especially those residing in informal settlements near flood-prone zones, are least able to recover from flood-induced losses due to limited savings and inadequate access to insurance or credit facilities. Their housing structures, often made of mud walls and thatched roofs, are easily destroyed, forcing repeated rebuilding that deepens poverty cycles. Youth engaged in casual labour and sand harvesting also experience recurring income shocks as floods disrupt riverine and agricultural production, further undermining household resilience. Conversely, a small segment of households with diversified income sources, such as salaried employment or established trade, demonstrate moderate coping ability and quicker recovery after flood events.

3.4.4 Human wildlife conflicts

Human–wildlife conflicts (HWC) have increasingly become a localized yet significant climate-related hazard in Mwarakaya Ward, particularly in areas bordering degraded forest ecosystems and wildlife corridors. Historically rare, these conflicts now occur approximately once every five to six years, with a notable surge in 2025 when several incidents were reported in Ondoni and Vwevwesi villages adjacent to Kaya Chonyi (Figure 16). The rising frequency of HWC events is closely linked to the degradation of natural habitats, driven by deforestation, agricultural expansion, and changing rainfall patterns that reduce water and forage availability within forest ecosystems. As a result, wildlife species such as baboons, monkeys and wild pigs increasingly move beyond forest boundaries into farmlands and settlements, causing damage to crops and property.

The degradation of Kaya Chonyi has intensified exposure to HWC by diminishing the natural barriers and food sources that once confined wildlife within these ecosystems. Reduced forest canopy cover and shrinking water sources have altered wildlife movement patterns, pushing animals into nearby agricultural zones. Similarly, the encroachment of agricultural activities in the forest has been subjected to wildlife damage. Vulnerability analysis indicate Vwevwesi (5) has the highest exposure followed by Ondoni (3) attributed to their close proximity to Kaya Chonyi (Figure 17). From a socio-demographic perspective, farmers living along Kaya Chonyi and pastoralists in open grazing areas are the most exposed groups. Women and children, who spend long hours on farms and at water collection points, face heightened physical danger from

wildlife encounters, while widows, elderly persons, and persons abled differently (PAD) are particularly vulnerable due to limited mobility and inability to construct deterrent fences or invest in alternative livelihoods. The destruction of crops and livestock not only threatens household food security but also forces families to depend on purchased food or relief support, thereby deepening poverty levels. Over time, HWC incidents have expanded in both frequency and severity, reflecting a broader ecological imbalance as wildlife habitats shrink and human settlements expand.

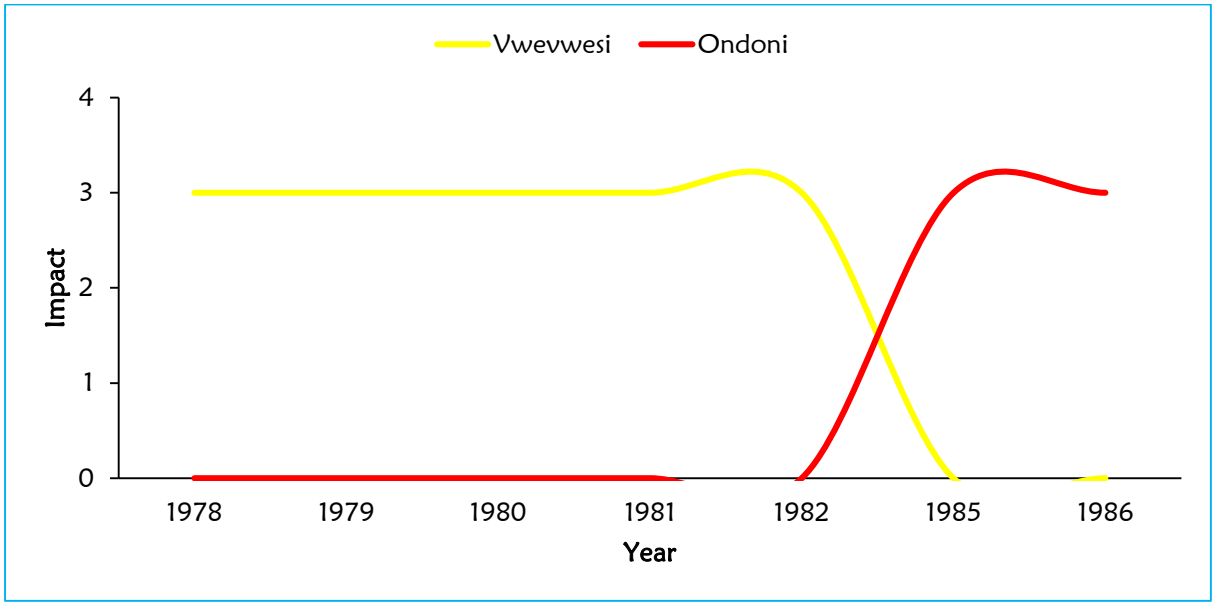

Figure 16: Human wildlife conflict prevalence historical trend (Source: PCVA, 2025)

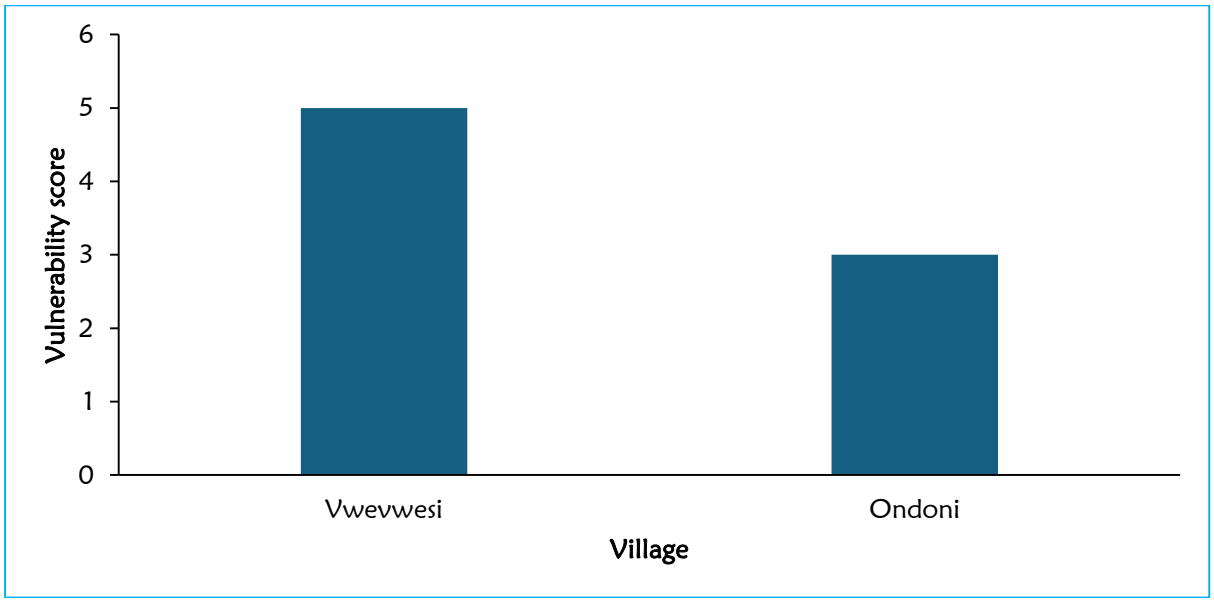

Figure 17: Human wildlife conflict vulnerability score in Mwarakaya Ward (Source: PVCA, 2025)

3.4.4.1 Exposure to Human–Wildlife Conflicts

Exposure to human–wildlife conflicts in Mwarakaya Ward is primarily concentrated in communities and ecosystems located near forested and degraded wildlife habitats, particularly around Vwevwesi and Ondoni Villages adjacent to Kaya Chonyi These areas experience the highest interaction between humans and wildlife due to diminishing vegetation cover, shrinking water sources, and expansion of farmlands into forest buffer zones. The degradation of Kaya

Chonyi Forest has significantly reduced natural barriers and food availability for wildlife, driving species such as wild pigs, baboons, and monkeys to forage in nearby farms. This exposes local farmers to frequent crop raids and destruction of farmlands, especially during dry seasons when natural forage is scarce.

The exposure extends to livestock keepers who graze animals near forest fringes as this increases the likelihood of predation and competition. Women and children, who spend extended hours on farms, in forest peripheries, or collecting water and firewood, are the most exposed to physical danger from wildlife encounters, while widows, the elderly, and persons abled differently (PAD) face heightened exposure due to limited mobility and inability to erect protective fences or respond quickly to threats.

3.4.4.2 Socio-Demographic Vulnerability to Human–Wildlife Conflicts (HWCs)

Socio-demographic vulnerability to human–wildlife conflicts in Mwarakaya Ward is shaped by livelihood type, gender, age, and proximity to wildlife habitats, with the highest vulnerability observed among resource-dependent and low-income households. Smallholder farmers and pastoralists residing near Kaya Chonyi Forest and along forest edges in Vwevwesi and Ondoni villages are the most affected, as their livelihoods rely heavily on land and water resources that overlap with wildlife ranges. Frequent crop raids by baboons, monkeys, and wild pigs, undermine household food security and income stability. These losses are especially devastating for poor households that lack access to compensation mechanisms or alternative livelihood options.

Women are disproportionately vulnerable because they perform daily activities that heighten their exposure, including farming, firewood collection, and water fetching near forested areas. The constant fear of wildlife encounters limits their mobility and reduces productivity in agriculture and trade. Children also face high vulnerability as they often accompany parents to farms or water sources, exposing them to physical danger. Widows, elderly persons, and persons abled differently (PAD) experience compounded vulnerability due to restricted mobility, limited ability to construct deterrent barriers, and dependence on external assistance for protection and recovery after incidents.

The economic ripple effects of human-wildlife conflicts (HWCs) in Mwarakaya Ward intensify vulnerability across the broader community. Crop and livestock destruction compel households to buy food, rely on relief aid, or migrate temporarily, disrupting education, social cohesion, and local markets. Over time, the growing frequency and severity of HWCs have deepened social inequalities. Wealthier households with stronger structures, fences, or alternative income sources recover faster, while poorer families remain caught in cycles of loss and indebtedness. Socio-demographic vulnerability to HWCs in the Ward underscores the intersection of environmental degradation, poverty, and gender inequality. This highlights the necessity for inclusive wildlife management strategies, gender-responsive compensation schemes, and community-led mitigation measures that protect the most vulnerable while promoting coexistence between people and wildlife

3.5 Classification of exposure and Resilience through livelihoods

The ward is characterised by livelihoods that are highly exposed to environmental, economic and institutional shocks, while its capacity for resilience remains constrained by limited income diversity, low margins and persistent poverty. The dominant cash-income sources are informal and closely tied to agriculture and transport services with modest supplementary incomes from micro-enterprises, livestock or poultry rearing. At the same time, county data indicate that Kilifi's poverty headcount has been around 46.1% per cent, higher than the national average.

These factors combine to yield a system with narrow buffers and high vulnerability, thereby demanding careful classification of exposure and resilience through the livelihood perspective.

3.5.1 Livelihood Structure and Monthly Incomes

Table 2 below presents typical monthly income figures (in KES) across four villages; Mbuyuni, Ondoni, Silala and Vwevwesi, illustrating the range of economic activities within these communities. Minning emerged as the highest income generator, with average monthly earnings of KES 50,000. Livestock production also contributes significantly, providing KES 10,000 to KES 30,000 per month, while agriculture offers a steady income averaging KES 20,000. Collectively these activities constitute the main income sources supporting livelihoods in the area.

Aggregate mining is a crucial economic activity that contributes significant income to local communities; however, it is vulnerable to climate-related changes such as flooding, leading to land degradation, erosion of riverbanks, water pollution, and habitat destruction, which result from unregulated and intensive extraction activities. This highlights the urgent need for responsible and sustainable management practices to mitigate environmental impacts and protect the livelihoods dependent on these ecosystems.

Agriculture, while also a key livelihood source, remains exposed to multiple vulnerabilities, including, unpredictable weather patterns, and pests and disease outbreaks. These risks lead to unstable agricultural production, which translates to unstable earnings for farmers, especially during hazard prone seasons.

In contrast, livelihoods such as small-scale trade, firewood sale and poultry rearing, yield significantly lower monthly incomes, typically ranging between KES 2,000 to 6,000. These activities are predominantly undertaken by women and youth and are characterized by limited capital investment, small market reach, and high sensitivity to seasonal variations. Consequently, households engaged primarily in these sectors exhibit low financial resilience, with minimal capacity for savings or reinvestment.

Table 2: Monthly earnings by livelihood and village (Source: PCVA, 2025)

Livelihood	Income (KES)			
	Mbuyuni	Ondoni	Silala	Vwevwesi
Aggregate mining	50,000	50,000	-	-
Livestock production	30,000	15,000	10,000	30,000
Fruit selling	-	50,000	-	-
Agriculture	15,000	25,000	20,000	20,000
Wine tapping	20,000	-	18,000	-
Motorbike 0perations	15,000	16,000	12,000	10,000
Sand harvesting	-	-	12,000	-
Selling herbs	-	15,000	-	20,000
Small scale businesses	-	5,000	-	5,000
Bee keeping	-	-	-	10,000
Firewood sale	-	-	-	6,000
Tree nurseries	-	-	-	20,000

The poverty profile of the villages reflects the economic fragility of its livelihoods. Field data show that many households in all four villages experienced prolonged periods without adequate income to meet their basic needs over the past year. Figure 18 below presents the number of people who reported lacking money to meet critical needs such as food, housing, clothing, transport, and medication in the last 12 months

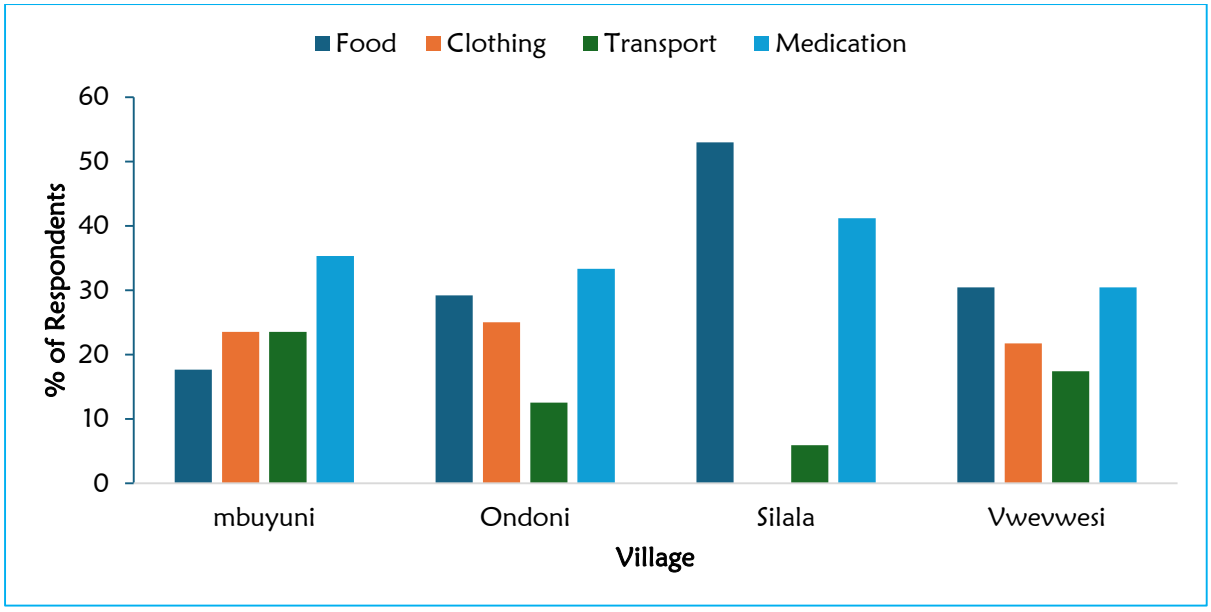

Figure 18: Respondents who lacked money to meet basic needs in the last 12 months (Source: PCVA, 2025)

This data provides evidence of widespread deprivation across the ward. The inability to afford basic necessities such as food and medication is moderately high, with 50% to 30% of households unable to meet these needs. Approximately, 30% to 35% of households per village, reported to lack funds for medical expenses reflecting persistent barriers to accessing affordable healthcare, increasing vulnerability to illness and reducing productivity. In addition, high numbers of people lacking transport and clothing reflect the broader structural poverty that limits both access to markets and participation in income-generating activities

3.5.2 Vulnerabilities of Livelihood Options

All livelihood options in the ward while essential for household sustenance, expose the community to varying degrees of risk. Sand harvesting, though offering the highest cash income, is impacted by fluctuations in river and water levels and sediment availability. Unsustainable sand harvesting combined with climate variability can lead to resource depletion, land degradation, and reduced water availability for domestic and agricultural use, threatening both the ecosystem and community livelihood.

Agriculture, predominantly rain-fed, faces direct risks from droughts and flooding, leading to crop failures and reduced yields. Variability in rainfall disrupts planting cycles and exacerbates pest and disease outbreaks, increasing food insecurity and income loss for rural farmers who rely on subsistence and small-scale farming.

Small scale trade activities, practiced mainly by women and youth, offer minimal financial returns and are constrained by limited capital, seasonal demand, and competition. Flooding and poor drainage often disrupt trading days, while low purchasing power within the local market restricts profitability. Additionally, livelihood activities such as wine tapping rely heavily on the health of palm trees, which are vulnerable to droughts and irregular rainfall. Similarly, beekeeping suffers risks from altered flowering seasons and decreased nectar availability caused by climate variability. Together, these factors reduce the resilience and income potential of small-scale traders and agricultural livelihoods.

Underlying all these vulnerabilities is the persistent issue of poverty. The low-income levels indicate that most households operate within or just above subsistence thresholds, leaving little

room for investment or shock recovery. In this context, minor external disturbances, such as, a spike in fuel costs, or an illness in the family can severely undermine household stability. The reliance on informal, low-profit activities without access to formal financial services amplifies this fragility.

3.5.3 Overall Classification and Implications

The cumulative evidence positions Mwarakaya Ward within a high exposure and low resilience category. The community's economic dependence on climate-sensitive and informal livelihoods, combined with high poverty levels and gendered unpaid work burdens, restricts adaptive capacity and heightens vulnerability. Addressing these constraints requires integrated measures that improve income stability in high-risk sectors, promote diversification into higher-value chains, reduce women's unpaid workload through labour-saving interventions, and strengthen social safety nets. Expanding access to microfinance, improving infrastructure such as roads and investing in ecosystem-based adaptation will be vital in shifting the ward's resilience profile from reactive coping to proactive adaptation.

3.6 Analyzing causes of vulnerability.

3.6.1 Classification of Severity of the Vulnerabilities and Their Causes

The ward exhibits varying degrees of vulnerability across its four sampled villages, Mbuyuni, Ondoni, Silala and Vwevwesi, largely determined by their ecological location, livelihood dependence, and adaptive capacity. The classification was informed by participatory scoring and spatial analysis. Vwevwesi and Ondoni demonstrated high vulnerability due to their proximity to Kaya Chonyi, dependence on forest resources for timber, herbal medicine, and exposure to human wildlife conflicts and drought. Mbuyuni and Silala were classified as moderately vulnerable, where drought, pest infestation, and limited water access threaten agricultural productivity.

The severity of vulnerability is compounded by limited livelihood diversification, chronic poverty, and inadequate institutional support. Communities relying on agriculture are more exposed to climatic fluctuations, facing recurrent crop failures. Women experience disproportionate vulnerability due to their engagement in low-return economic activities, restricted access to credit and training, and the intensification of unpaid domestic work during climate shocks. This classification underscores the multidimensional nature of vulnerability, where physical exposure, socio-economic status, and gender intersect to determine the overall risk profile.

3.6.2 Identification of Unsafe Conditions

Unsafe conditions in the ward refer to the immediate settings and circumstances that heighten exposure to climate-related hazards. Settlements located along riverine ecosystems in Ondoni face recurrent flooding resulting in frequent loss of assets and disruption of livelihoods

From a gender perspective, unsafe conditions are further reflected in the burden placed on women who must travel longer distances to collect water and firewood during dry seasons, often through degraded landscapes that pose physical risks. Limited access to climate-resilient infrastructure such as raised storage facilities, intensifies vulnerability for both men and women, but particularly for households headed by widows or persons abled differently.

3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability

Dynamic pressures represent the intermediate processes through which underlying socio-economic and institutional weaknesses are translated into direct, unsafe conditions that heighten vulnerability. In Mwarakaya Ward, these pressures manifest through rapid population growth, unregulated land-use change, overexploitation of natural resources, and weak

enforcement of environmental by-laws, all of which interact to degrade ecosystems and undermine resilience.

Kaya Chonyi, one of the ward's key ecological assets, covering approximately 391.82 hectares, was found to have 26.91 hectares (22.58%) of its area degraded mainly due to encroachment by local communities, driven by farming and informal settlement expansion. Despite having a well-structured, locally led management system, the forest remains highly vulnerable to a range of human-induced threats. Persistent conflicts over forest boundaries continue to strain community relations, undermining collective conservation efforts. Local accounts indicate that some individuals encroach into the forest during prolonged droughts to establish temporary farms, while others have illegally occupied portions of forest land for construction. Increasing population pressure and high unemployment levels have further intensified the exploitation of forest resources. Moreover, unresolved land ownership disputes and limited community cohesion pose additional challenges to the forest's preservation factors that, if left unaddressed, could ultimately result in the complete loss of this vital ecosystem.

The primary drivers of this decline are extensive livestock grazing, which impedes natural regeneration through trampling and browsing of seedlings, and unsustainable tree harvesting for construction materials and fuelwood. These pressures indicate persistent livelihood-related dependencies on forest resources and inadequate regulation of extraction activities. The degradation of Kaya Chonyi not only reduces vegetation cover and biodiversity but also compromises its ecosystem services, such as soil stabilization, microclimate regulation, and carbon sequestration, thus amplifying community exposure to droughts, strong winds, and floods.

From an institutional perspective, weak coordination between county departments, local community structures, and national agencies constrains effective resource governance. Enforcement of conservation by-laws remains sporadic, while early warning systems for floods and droughts are either absent or poorly operationalized, reducing preparedness and timely response. Economically, fluctuating market prices for seaweed and limited access to credit facilities continue to undermine household income stability. On the social front, the exclusion of women from decision-making forums, particularly in forest and fisheries management committees, limits the inclusivity and sustainability of adaptation initiatives.

3.6.4 Identification of Underlying Causes of Vulnerability

The underlying causes of vulnerability in Mwarakaya Ward stem from interlinked socio-economic, environmental, and institutional deficiencies that collectively erode household resilience and adaptive capacity. Structural poverty remains pervasive, limiting access to education, technology, financial services, and diversified livelihood opportunities. According to the Participatory Climate Vulnerability Assessment (PCVA, 2025), the average household size in Mwarakaya Ward is 7 persons per household, indicating a high dependency ratio and increased pressure on limited natural and financial resources. Larger households require greater quantities of food, water, and energy, yet often rely on subsistence livelihoods and ecosystem-dependent activities such as medicinal plants, and small-scale agriculture. This dependency entrenches poverty cycles and increases exposure to environmental and economic shocks.

Environmental degradation constitutes another major underlying factor, manifesting in Kaya Chonyi and along riverine ecosystems. The loss of vegetative cover reduces the capacity of ecosystems to buffer against climate-induced hazards, particularly floods, droughts, and strong winds, thereby increasing community exposure and reducing long-term ecological stability.

Institutionally, weak governance frameworks and limited coordination between Kilifi County Government, the Ward Climate Change Planning Committee (WCCPC), and relevant national agencies hinder the implementation of adaptive interventions. Resource constraints, overlapping mandates, and inadequate policy enforcement reduce the effectiveness of climate action at both ward and county levels. Furthermore, the absence of comprehensive early warning and information dissemination systems results in delayed community responses to climatic hazards, shifting the focus from preparedness to crisis management

Gender inequality further compounds vulnerability. Patriarchal norms and customary land tenure systems restrict women's ownership and control of productive assets, constraining their participation in decision-making and their capacity to adopt climate-smart technologies or access credit facilities. Consequently, women remain overrepresented in low-return and climate-sensitive livelihood activities, limiting their economic independence and adaptive potential.

3.6.5 Prioritization of Causes

Prioritization of the identified causes was achieved through participatory ranking during the PCVA exercises, considering frequency, severity, and reversibility of impacts. The highest priority causes were poverty, environmental degradation, and weak institutional support systems. Poverty restricts adaptive investments and perpetuates reliance on climate-sensitive livelihoods, while environmental degradation reduces the ecosystem's regenerative capacity. Weak institutional mechanisms delay response and hinder the implementation of ward-level adaptation initiatives.

Medium-priority causes include gender inequality, population pressure, and lack of access to climate information, all of which significantly influence vulnerability dynamics. Addressing these priority causes requires an integrated approach, strengthening governance structures, enhancing livelihood diversification, and promoting gender-responsive adaptation strategies. Interventions such as forest restoration, provision of microcredit for women's enterprises, and establishment of community-based early warning systems are essential to reduce long-term vulnerability and build resilience across all villages in the ward.

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards

Gender and age critically shape patterns of exposure, sensitivity, and adaptive capacity across population groups in Mwarakaya Ward, where the dominant climate hazards; drought, floods, pests and diseases, and human–wildlife conflicts arising from wildlife movement within Kaya Chonyi, interact with entrenched socio-cultural inequalities and chronic poverty (Table 3). These hazards disproportionately affect women, girls, youth, elderly persons, and Persons Abled Differently (PADs), whose vulnerability is determined by restricted access to productive assets, unequal household decision-making power, gendered labour divisions, and mobility limitations (UN Women, 2022; IPCC, 2022).

A major driver of vulnerability among women and girls in Mwarakaya is the heavy and undervalued burden of unpaid domestic and care work, including childcare, cooking, water fetching, firewood collection, elder care, cleaning, and subsistence farming. During PCVAs, this work was valued at KES1,500 per day, yet remains unrecognized within formal economic systems. Nationally, women undertake over 70% of unpaid care labour (KNBS, 2023; UNDP, 2022). Climate hazards intensify this workload by increasing the time required to secure household water, food, and fuelwood, thereby deepening "time poverty" and limiting women's participation in income generation, leadership spaces, capacity-building initiatives, and climate adaptation planning (ILO, 2018; UN Women, 2023).

Intra-household power imbalances in Mwarakaya further compound vulnerability. Many women lack decision-making authority regarding land use, agricultural inputs, income allocation, and children's schooling. This limits their ability to adopt climate-resilient practices and access extension services or credit. Climate stress often heightens intra-household tensions and contributes to gender-based violence (GBV), including emotional, physical, and economic abuse, which undermines women's autonomy and mobility. Kenya's rising cases of femicide exacerbate fear among women and girls, restricting movement, reducing participation in markets and training, and weakening adaptive capacity (UNFPA, 2023; Kenya Police Service, 2024).

Drought is one of the most severe hazards in Mwarakaya Ward, contributing to chronic water scarcity, reduced agricultural productivity, livestock stress, and widespread household food insecurity. Women and girls walk long distances, often several kilometres, to access water and firewood, exposing them to dehydration, musculoskeletal strain, heat stress, and protection risks such as harassment (UNICEF, 2022). Female-headed households, which are common due to male out-migration and widowhood, experience heightened vulnerability due to limited labour and financial resources. Men face significant losses in crop yields and livestock productivity, increasing economic stress and reducing their capacity to support household needs. Youth (18–35) experience declining casual labour opportunities and limited apprenticeship prospects, contributing to idleness and increased exposure to unsafe coping strategies (UNFPA, 2022). Elderly persons and PADs face severe challenges accessing water and food due to mobility restrictions, increasing dependency on family members (WHO, 2021).

Flooding affects low-lying and riverine areas of Mwarakaya, damaging homes, local markets, feeder roads, and WASH infrastructure. Women, who dominate informal trade and household management, suffer reduced income, increased caregiving duties, and heightened stress as they manage displaced family members and contaminated water supplies. Men lose daily wage labour opportunities, including small-scale construction and transport services, while damage to agricultural fields reduces income stability. Youth experience prolonged school interruptions, high idleness, and vulnerability to exploitation during displacement (UNICEF, 2021). Elderly persons and PADs face extreme difficulty evacuating due to limited mobility and dependence

on caregivers, increasing their exposure to disease and neglect in crowded shelters (WHO, 2020).

Pests and livestock/crop diseases significantly undermine agricultural productivity in Mwarakaya Ward, where most households rely on maize, cassava, legumes, and poultry. Women, who manage a large share of subsistence farming and small livestock, experience reduced yields, post-harvest losses, and declining household food security. Their adaptive capacity is constrained by limited access to certified seeds, pest-control inputs, veterinary services, and agricultural extension (FAO, 2021). Men incur financial losses from declining crop production and livestock morbidity. Youth lose agricultural casual labour opportunities and income from small enterprise activities. Elderly persons and PADs face heightened food insecurity as households prioritise feeding younger and more productive members.

Human–wildlife conflict is a major hazard in Mwarakaya, particularly for households living near Kaya Chonyi, where wildlife, including baboons, monkeys and bushpigs, invades farms, destroys crops, and threatens safety. Women and girls restrict movement due to fear of wildlife encounters, limiting access to farms, water sources, firewood areas, and local markets. Limited decision-making power and documentation challenges further restrict women's access to compensation schemes. Men face higher risks guarding farms at night and suffer significant crop losses, increasing household vulnerability. Youth assisting in farm protection face physical danger and disrupted schooling. Elderly persons and PADs are extremely vulnerable during wildlife incidents due to slow mobility and an inability to escape quickly, heightening dependency and emotional stress (UNEP, 2022).

Gender-based violence (GBV) further compounds these gendered vulnerabilities, functioning both as a climate-induced risk and as a structural barrier to resilience in Pongwe Kikoneni Ward. Emerging global evidence demonstrates a strong association between climate stressors and rising GBV incidence, with every 1°C increase in temperature linked to a 4.7% rise in intimate partner violence (UN Spotlight Initiative, 2024). Heatwaves have been associated with a 28% increase in femicide, while climate-driven displacement heightens risks of child marriage, sexual exploitation, and trafficking (UN Spotlight Initiative, 2024). In Kenya, rising temperatures and socioeconomic stressors have coincided with escalating femicides with over 930 cases documented between 2016 and 2024, highlighting how climate stress intersects with entrenched gender inequalities (Kenya Femicide Data Study, 2024). These dynamics are most acute in fragile, resource-dependent communities such as Pongwe Kikoneni, where drought, floods, and livelihood loss amplify household tensions, reduce economic security, and disproportionately expose women and girls to physical, psychological, and sexual violence. GBV undermines climate resilience by restricting women's mobility, decision-making autonomy, access to productive assets, and participation in ward-level adaptation processes (UN Women, 2022). As such, GBV must be integrated into anticipatory climate action, beyond response and preparedness, through early warning systems, forecast-based triggers, community surveillance, and pre-positioned support services coordinated with mandated county structures such as the County Gender Department, Social Services, child protection units, and GBV coordination mechanisms that programmes like COSME have actively engaged. Embedding GBV mitigation within resilience planning ensures that climate adaptation measures safeguard rights, safety, and justice, which are foundational to sustainable and inclusive resilience building.

Table 3: Gender/Age Vulnerability Analysis for Climate Hazards

Women	Men	Youth	Elderly	PADs
Drought				
• Heavy workload collecting water and firewood as sources dry up • Walk long distances (3–5 km) → heat stress, dehydration & protection risks • Reduced time for childcare and income activities • Psychological stress due to food and water shortages	• Livelihood insecurity from reduced crop yields, livestock productivity, and forest resources • Increased financial pressure and stress • Reduced availability of agricultural labour opportunities	• Reduced productivity and income from casual labour losses • High idleness → increased risk of crime, drugs, and unsafe coping strategies • Higher incidence of early marriages and pregnancies among girls	• Mobility limitations restrict access to water and food • Increased susceptibility to heat stress, dehydration, and hunger • Dependence on caregivers worsens during drought periods	• High dependency for water, food, and healthcare access • Mobility constraints limit ability to queue for relief supplies • Increased risk of neglect during prolonged shortages
Floods				
• Increased caregiving burdens for sick or displaced family members • Loss of income from informal trade due to market inaccessibility • Contaminated wells force longer distances to find safe water • Elevated psychosocial stress	• Loss of income from transport, construction & farm labour • Increased stress from livelihood losses	• School closures disrupt learning and training • Loss of casual labour - increased idleness, crime, and drug use • Girls particularly vulnerable to early marriage and pregnancy	• High susceptibility to waterborne diseases and cold exposure • Limited mobility restricts evacuation • Greater dependence on others for shelter, food, and medicine	• Severe mobility barriers during evacuation and accessing aid • High risk of injury and neglect in shelters • Limited access to emergency services

4.1 Children's Vulnerability to Climate Hazards and Key Mitigation Measures

Children in Mwarakaya Ward are among the most climate-vulnerable groups due to their rapid physiological development, high dependency on caregivers, and limited capacity to independently anticipate or cope with environmental hazards (UNICEF, 2022). Their vulnerability is heightened by chronic poverty, malnutrition risks, reduced WASH access, and recurrent disruptions to schooling, all of which are aggravated by the ward's major climate hazards; drought, floods, pests and diseases, and human–wildlife conflicts from Kaya Chonyi.

Drought exposes children to acute malnutrition, stunting, anaemia, and dehydration as household food production declines and water scarcity worsens (WHO, 2020). Many children, especially girls are required to participate in water collection, walking long distances under high temperatures, which increases exposure to heat stress, exhaustion, and protection risks such as harassment (UNICEF, 2022). These responsibilities reduce school attendance and hinder learning outcomes (UNESCO, 2021).

Floods damage homes, disrupt learning environments, and destroy WASH facilities, increasing the risk of drowning, diarrhoeal diseases, cholera, respiratory infections, and skin diseases among children (WHO, 2020). When schools are used as temporary evacuation centres, learning cycles are interrupted, school feeding programmes are suspended, and children experience emotional distress linked to displacement (UNICEF, 2021). Flood-related instability heightens risks of family separation, anxiety, behavioural changes, and sleep disturbances among children (UNFPA, 2021).

Pests and diseases, including crop infestation and livestock illness, undermine household food security, forcing families to reduce meal frequency or rely on less nutritious foods. This exposes children to micronutrient deficiencies, compromised immunity, and delayed development (FAO, 2021). Children from food-insecure households show higher rates of school absenteeism and reduced cognitive performance (UNDP, 2022).

Human–wildlife conflicts (HWC) from Kaya Chonyi pose additional hazards, as children often accompany caregivers to farms, water points, or firewood areas where wildlife, particularly baboons, monkeys, bushpigs, and occasionally snakes, may pose threats (UNEP, 2022). HWC heightens risks of physical harm, emotional trauma, and reduced mobility, as families restrict children's movement during peak wildlife activity. Fear of wildlife encounters also disrupts schooling, play, and participation in outdoor activities essential for healthy development.

Climate-related livelihood losses and stress in Mwarakaya increase the likelihood of harmful coping strategies, including early marriage, child labour, transactional behaviour among adolescents, and withdrawal from school, especially during prolonged drought or after severe flood events (UNFPA, 2022). These practices violate child protection norms and perpetuate cycles of poverty and vulnerability.

Protecting children in Mwarakaya Ward requires child-centred climate adaptation measures, including strengthening community- and school-based early warning systems (UNDRR, 2020), enhancing nutrition surveillance systems to trigger early action during drought (UNICEF, 2022), and establishing climate-resilient WASH facilities such as protected boreholes, elevated latrines, and rainwater harvesting systems (WHO, 2020). Flood-safe, climate-proof education infrastructure with storm-resistant roofing and elevated floors is essential to maintain learning continuity (UNESCO, 2021).

Equally important is the provision of psychosocial support, including child-friendly spaces, trauma-informed counselling, and family-tracing mechanisms during displacement (UNFPA,

2021). Strengthening social protection mechanisms, such as school feeding, bursaries, and emergency child grants, helps households avoid negative coping strategies during climate shocks (UNDP, 2022). Finally, empowering teachers, community health volunteers (CHVs), and local leaders to strengthen child protection structures ensures children's safety, dignity, and well-being before, during, and after climate hazards.

5 Chapter 5: Adaptation Actions

5.1 Introduction

Building on the findings of the hazard and vulnerability assessment, this chapter presents the adaptation actions developed for Mwarakaya Ward. These interventions were formulated through participatory consultations using PCVA tools and impact chain analysis, ensuring they accurately reflect the ward's environmental characteristics and socio-economic conditions. The proposed actions address both immediate climate-related stresses such as drought, floods, Pests and diseases and human wildlife conflicts and the underlying structural vulnerabilities related to poverty, ecosystem degradation and institutional capacity gaps. that affect community livelihoods and ecosystem health. They are designed to strengthen long-term resilience across sectors through a combination of short-term, medium-term, and long-term strategies.

The proposed measures encompass short, medium, and long-term strategies. Short-term actions include community awareness creation, institutional strengthening, and capacity-building initiatives. Medium and long-term interventions involve climate-resilient infrastructure development, ecosystem restoration, and the promotion of nature-based and technological solutions. For instance, the installation of water harvesting and storage systems is intended to mitigate drought, while forest restoration will reduce the impacts of flooding,

The formulation of these adaptation measures integrates local and indigenous knowledge with scientifically validated best practices drawn from Kenya's National Adaptation Plan (2015–2030), the National Climate Change Action Plan (2023–2027), and the Kilifi County Climate Change Action Plan (2023–2027). Gender equity and social inclusion principles are put in place throughout to ensure that adaptation benefits are distributed equitably. Women who are often disproportionately affected by water scarcity and declining agricultural productivity are targeted through livelihood diversification, leadership, and financial empowerment initiatives. Meanwhile, youth are engaged through innovation-driven adaptation activities such as climate information dissemination, ecosystem monitoring and innovation in sustainable resource use.

Currently, there are ongoing community-led projects in Mwarakaya Ward that complement these adaptation priorities, including, bee keeping, tree planting initiatives, and local savings and credit groups (VSLAs) that support climate-resilient livelihoods. These initiatives demonstrate the community's commitment to sustainable adaptation and provide a strong foundation upon which the proposed actions will build.

Implementation of the adaptation measures will occur at multiple levels i.e., household, community, ward, county, and national, under the coordination of the Ward Climate Change Planning Committee (WCCPC). Collaborative engagement among local organizations, CFAs, WRUAs, and relevant government institutions will be essential to ensure coherent, inclusive, and sustainable delivery of the adaptation agenda across Mwarakaya Ward

5.2 Existing climate-resilient and livelihood-enhancing projects

Mwarakaya Ward currently benefits from a diverse network of community-led initiatives, institutional partners, and development organizations implementing climate-resilient and livelihood-enhancing projects (Table 4). These interventions collectively strengthen local adaptive capacity, promote ecosystem restoration, and support sustainable resource use. Existing strategies focus on forest restoration, water access, sustainable agriculture, and community-based financial inclusion, demonstrating strong collaboration between the community, civil society, and county agencies.

Key ongoing initiatives include the establishment and maintenance of tree nurseries and reforestation programs led by groups such as Botanic Gardens Conservation International, Kuna Nuru Group, Ondoni Conservation Group, Tsetsezani Group, and Kaya Chonyi Conservation Group, with technical and financial support from Plan International. These efforts contribute to catchment restoration, soil conservation, and increased vegetation cover under the broader county ecosystem restoration strategy.

Table 4: Existing climate resilient and livelihood enhancing projects in Mwarakaya Ward

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
Tree Nursery and Tree Planting	Botanic Gardens Conservation International,	Ondoni, Vwevwesi,	Environmental Restoration	Supports afforestation and reforestation under the Kilifi County Forest and Landscape Restoration Strategy; enhances vegetation cover and soil stability.
Table Banking and VSLA Support	Plan International	Vwevwesi, Silala, Ondoni,	Social Protection / Financial Inclusion	Strengthens household savings and access to credit; enhances adaptive capacity through women and youth-led savings groups.
Bee keeping	Financing Locally Led Climate Action (FLLoCA) Program	Vwevwesi	Livelihood diversification	Supports diversification of livelihoods with sustainable alternatives
Climate smart agriculture	Empowering Farmers Foundation (EFF)	Vwevwesi, Mbuyuni	Food security	Enhances farmer input availability for diversified agricultural production

5.3 Adaptation actions

Adaptation actions in Mwarakaya Ward are designed as an integrated and gender-responsive climate resilience framework that protects livelihoods, strengthens ecosystems, and safeguards the well-being of all population groups across the four villages; Mbuyuni, Ondoni, Silala and Vwevwesi. (Table 5). These actions respond to six priority climate hazards; drought, floods, pests and diseases, and human–wildlife conflicts and are anchored in the Kenya Climate Change Act (2023), the National Adaptation Plan (2015–2030), the Kilifi County Climate Change Act (2021), and national frameworks on gender equality, child protection, and sexual and reproductive health and rights (SRHR). Implementation follows a phased approach comprising short-term interventions (≤1 year), medium-term measures (≤3 years), and long-term interventions (>3 years), all anchored in gender equity, social inclusion, SRHR continuity, and community ownership to ensure climate benefits are distributed fairly and sustainably.

Under Thematic Area One, Water Security and Management focus on strengthening household, institutional and community access to reliable and safe water resources across Mbuyuni, Ondoni, Silala and Vwevwesi villages. In the short term, households will prioritize rainwater harvesting through the installation of gutters and storage tanks at an estimated cost of KES 50,000 per household, reducing the time burden on women and girls who undertake most water collection. Immediate ecosystem rehabilitation will be undertaken through fencing of springs in Vwevwesi Village, budgeted at KES 100,000, thereby restoring water storage capacity for domestic and livestock use. These short-term actions are complemented by sensitization of women and youth on household water treatment practices, implemented through barazas facilitated by CHPs, NGOs and KIMAWASCO at KES 20,000 per village. Medium-term measures focus on institutional strengthening through the installation of 10 m^3 high-capacity water tanks and rooftop harvesting systems in one school per village, costed at KES 400,000 per institution. These facilities will support school WASH systems and provide fallback water supplies during drought. The programme will include installation of solar powered water pumps on existing wells in Ondoni and Vwevwesi villages to reduce women workload, each budgeted at KES 70,000. These interventions will be delivered collaboratively through Kilifi County Government (FLLoCA), NGOs, private sector partners and KIMAWASCO.

Long-term actions advance strategic water supply expansion through drilling and solarizing one borehole in Mbuyuni, Ondoni, Vwevwes villages, each budgeted at KES 5,000,000, to ensure perennial water availability for domestic and small-scale irrigation use. To further enhance resilience, the construction of additional water pans in Mbuyuni, Ondoni and Vwevwesi at KES 3,000,000 per water pan will store excess rainfall and provide year-round supply for irrigation and livestock. These long-term interventions collectively aim to reduce climate-related water stress while improving gender-equitable access to safe water.

Thematic Area Two focuses on strengthening household food security and ecosystem restoration and livelihoods through climate-smart agriculture and resilience design. In the short term, women and youth will be capacitated on resilience design techniques, soil fertility enhancement, water retention using zai pits, mulching and efficient farm layout, delivered through county extension services and NGOs at KES 10,000 per farmer per 12-month cycle. Training will also cover climate-smart agriculture, including drought-tolerant crop varieties (cassava, katumani maize, sorghum), hay and silage making, as well as sensitization on index-based crop and livestock insurance. Medium-term interventions include distributing certified seeds, fertilizers, pesticides and livestock inputs to trained groups in all four villages. Production support is costed at KES 55,000 per acre for maize, KES 40,000 per acre for cassava, and KES 10,000 per livestock farmer. Select women- and youth-led farms will be supported to fully implement resilient design techniques, forming demonstration farms for continuous peer learning. Establishing a Ward-level Farmer Field School (FFS) will institutionalize resilient

farming practices and ensure continuous technical support. Long-term actions prioritize market strengthening by aggregating produce from women and youth to enhance bargaining power and value addition. Livestock diversification will be supported through pasture establishment at KES 55,000 per acre, creation of fodder seed banks, and distribution of Gala goats (KES 15,000 each) and dairy cows (KES 100,000 each). These interventions reduce climate sensitivity while improving livelihood stability.

In addition, the area focuses on restoring degraded ecosystems and enhancing nature-based solutions. In the short term, women, youth and PADs will be trained to establish nurseries for priority tree species for agroforestry, boundary planting and woodlot establishment. Training is budgeted at KES 5,000 per trainee per day, and each nursery will cost KES 75,000 to establish. Training on briquette production will also be provided to reduce dependence on charcoal and mitigate forest degradation. Further measures will involve preparing a Conservation and Management Plan for Kaya Chonyi and building the capacity of Kaya elders to support implementation budgeted at KES 500,000.

Medium-term interventions include providing households with 100 priority seedlings per home at KES 5,000 per household, rehabilitating degraded sections of Kaya Chonyi at KES 150,000 per acre, provision of energy-saving jikos (KES 3,000) and kilns (KES 7,000) will reduce pressure on forest resources, enhancing long-term sustainability as well as of harvesting sand along Rivers Mwanzae and Mwabanda, implemented by NEMA, WRA and the County Government of Kilifi. Sensitization barazas at KES 20,000 per village will promote sustainable sand harvesting, riverbank protection and flood mitigation, ensuring ecological integrity of the watershed. Long-term actions emphasize construction of gabions and check dams along River Mwanzae, Mwabanda and eroded sections to reduce flood velocity and stabilize banks in Mbuyuni, Ondoni and Silala Villages, implemented by NEMA, WRA, Kilifi County Government, NGOs, Private Sectors.

Thematic Area Three strengthens adaptive capacity through diversified income streams and social safety nets. Short-term actions include training women, youth and PADs in alternative livelihoods such as kitchen gardening, beekeeping, poultry production and briquette making, delivered at KES 5,000 per trainee per day. VSLA governance structures will be strengthened to promote financial literacy and enhance adaptive capacity. Medium-term actions include provision of livelihood inputs, seeds and organic fertilizers, modern beehives (KES 7,000) and egg incubators (KES 100,000), to facilitate enterprise establishment. VSLAs will be supported to invest excess savings in green financial products such as Safaricom's Green Bond. Long-term actions include constructing a cultural centre and establishment of a nature trail in to promote tourism, as well as installation of solar power electric fence around Kaya Chonyi to mitigate human wildlife conflicts. The construction of a cultural center will be guided by a BOQ while the electric fence is budgeted at KES 7,000,000. The implementing parties will include KWS, NMK, Kilifi County Government, NGOs, Private Sectors.

Thematic Area Four addresses structural vulnerabilities to floods, strong winds and drought. Short-term actions include training community emergency response teams across all four villages at KES 5,000 per trainee. These foundational measures strengthen preparedness and immediate response to climate hazards. Medium-term interventions include a food-for-work programme, to engage youth in restoring drainage channels, cleaning culverts, constructing gabions and undertaking tree planting for erosion control. Early Warning Systems (EWS) will be developed and operationalized in all villages at KES 70,000 per village, improving access to predictive information on floods, droughts, and pest outbreaks. Community emergency response teams will be equipped with PPEs and first-aid kits which will be guided by BOQ. Long-term actions involve constructing raised footbridges and culverts at flood-prone crossing points in Mbuyuni,

Ondoni and Silala villages, and improving road drainage systems through the installation of new culverts to channel stormwater safely. All long-term infrastructural works will be guided by BOQs and implemented in partnership with Kilifi County Government, KERRA, KMD, NDMA, and NGOs.

Thematic Area Five addresses rising human–wildlife conflicts (HWC) especially in Ondoni and Vwevwesi adjacent to Kaya Chonyi. Short-term actions include community sensitization on conflict hotspots, safe behaviour to reduce encounters, and reporting mechanisms to KWS through the emergency hotline. Each sensitization forum will cost KES 20,000 per village. Long-term strategies involve installation of solar power electric fence around Kaya Chonyi, budgeted at KES 7,000,000.

Thematic Area Six centers on integrating gender equality, youth empowerment and inclusion of PADs into climate adaptation. In the short term, targeted GBV anticipatory actions will be developed for drought and floods, ensuring safe access to services for women and girls. Existing women's associations such as VSLAs will be used to disseminate information on rights, helplines and GBV response systems at KES 30,000 per VSLA. Youth boot camps addressing drug abuse, reproductive health and life skills will be implemented at KES 100,000 per camp, and children and youth will be vaccinated at local health facilities to reduce outbreak risks during climate shocks.

Medium-term actions include assessing and refining Early Warning Systems for climate-linked GBV anticipatory responses, and strengthening inclusive governance through training women, youth and PADs at KES 5,000 per person per day. GBV response mechanisms, including transfer vehicles and specialized first-aid and rape kits, will be procured based on BOQs. School feeding and scholarship support will be anchored on needs assessments to safeguard climate-vulnerable learners. Long-term interventions include developing and implementing a comprehensive Ward GBV anticipatory action plan (costed at KES 1,000,000 for consultancy services), constructing a county-level drug rehabilitation centre, and facilitating PADs to access mobility devices, assistive technology and entrepreneurial training through stakeholder engagement forums budgeted at KES 50,000.

Table 5: Mwarakaya Ward Adaptation Actions

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
Thematic area 1: Water Security and Management				
Rainwater harvesting through Installation of gutters on house roofs and water tanks at household level in Mbuyuni, Ondoni, Vwevwesi villages	Rainwater harvesting through installation of gutters and high-capacity water tanks (10m^3) in schools– one school in each village to enhance water access by women and children	Drilling and solarizing boreholes in Mbuyuni, Ondoni, Vwevwesi villages- one in each village	Kilifi County Government-FLLoCA, NGOs and Private Sector	50,000/HH, 400,000/School, 5,000,000/borehole
Fencing of springs in Vwevwesi village	Strategic (where embankment for water intake is viable) installation of solar powered water pumps on existing wells to reduce women workload (turn-around time) in Ondoni and Vwevwesi villages	Construction of water pans in Mbuyuni, Ondoni and Vwevwesi villages Construction of a sand dam along River Mwabanda in Ondoni village		100,000 per spring (Fencing), 70,000/ water pump, 3,000,000/water pan, 1,000,000/sand dam
Sensitizing women and youth on water treatment in Mbuyuni, Ondoni, Silala, and Vwevwesi villages			KIMAWASCO, Kilifi County Government, CHPs, NGOs and Private Sector	20,000 village baraza
Thematic area 2: Food security, ecosystem restoration and livelihoods				
Training and supporting women and youth on Resilience Design techniques from homesteads to watershed level (Soil fertility improvement, water conservation/retention (e.g. Zai Pits), farm design and use of local inputs to reduce farming costs)	Supporting select women and youth lead farmers to fully implement resilient design techniques including climate smart agriculture to serve as demonstration farms for continuing learning by other farmers to enhance sustainability of the interventions	Establishing a Farmer Field School in the Ward	Kilifi County Government, KARLO, NGOs and Private Sector	Farmer Field Schoos (FFFs) 10,000 per farmer for a 12-month cycle
Training women and youth on Climate Smart Agriculture (CSA) (drought tolerant crops such as katumani maize cassava, sunflower, groundnuts, cashewnuts, sorghum),	Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and extension services across Mbuyuni, Ondoni, Silala and Vwevwesi to trained women and youth groups	Aggregating women and youth farmers' produce to enhance farm produce marketability and enhance value addition and agribusiness opportunities		Production costs: Maize 55,000 per acre and Cassava 40,000 per acre

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
silage/hay making to address livestock feed challenges during drought in Mbuyuni, Ondoni, Silala, and Vwevwesi villages		Training women, youth and PADs on cassava value addition in Mbuyuni, Ondoni, Silala, and Vwevwesi villages		10,000 for livestock farmers per year or crop cycle (Free if it is provided by the government)
Sensitize and facilitate access to information on crop and livestock insurance (index-based drought coverage) targeting women and youth groups involved in CSA	Establish livestock fodder seed banks, pastures, and introduce drought-tolerant livestock breeds e.g., Gala goats/Dairy cows for women and youth groups in Mbuyuni, Ondoni, Silala, and Vwevwesi villages			Village barazas at 20,000, 55,000 per acre for pasture, 15,000 per Gala goat, 100,000 per dairy cow
Facilitate women and youth to establish tree nursery of priority species (Annex 1) and train them on on-farm tree production (Agroforestry, woodlots and boundary planting) in Mbuyuni, Ondoni, Silala, and Vwevwesi villages			Kilifi County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per day per trainee (50 pax from the five villages) 75,000 per nursery
Prepare a Conservation and Management Plan for Kaya Chonyi and build the capacity of Kaya elders to support implementation	Rehabilitate degraded areas in Kaya Chonyi		KFS, KEFRI, Kilifi County Government, CFAs, NGOs, Private Sector	500,000 for Plan Preparation and capacity building 150,000 per acre for rehabilitation 7,000,000 for fencing
Training women, youth and PADs on making organic briquettes (coconut husks, cassava peels) in Mbuyuni,	Provision of energy saving jikos and kiln across Mbuyuni, Ondoni, Silala, and Vwevwesi villages			5,000 per day per trainee (50 pax from the five villages)

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
Ondoni, Silala, and Vwevwesi villages				3,000 energy saving jiko & 7,000 per Kiln
Sensitization on riverbanks management and flood control and sand harvesting impacts along Rivers Mwanzae and Mwabanda	Regulation of harvesting sand along Rivers in Silala	Construct gabions and check dams along River Mwanzae, Mwabanda and eroded sections to reduce flood velocity and stabilize banks in Mbuyuni, Ondoni and Silala Villages	NEMA, WRA, Kilifi County Government, NGOs, Private Sector	20,000 for village barazas
Thematic area 3: Diversified Livelihoods and Social Protection				
Training women, youth and PADs on alternative livelihoods such as beekeeping, poultry farming and ecotourism (Kaya Chonyi and Mawe Meru) in Mbuyuni, Ondoni, Silala, and Vwevwesi villages	Provision of alternative livelihoods establishment materials/Equipment; modern beehives, incubators (Mbuyuni, Ondoni, Silala, and Vwevwesi villages	Construction of a cultural centre and establishment of a nature trail in Kaya Chonyi to promote tourism	Kilifi County Government, NMK, NGOs, Private Sector	5,000 per day per trainee (40 pax from the four villages) 7,000-Modern beehive, 100,000-Incubator & 30,000 per signage
Promote basketry and mat weaving (already existing) through providing market linkage initiatives in Vwevwesi and Silala (exhibition)			NMK, Kilifi County Government, NGOs, Private Sector	200,000-exhibition
Train VSLAs on Governance and alternative regenerative investment opportunities such as green bonds and money market funds to diversify interest income from surplus savings	Promote VSLAs excess savings investments in reputable interest earning schemes such as the green bond by Safaricom		County Government of Kilifi, NGOs, Private Sector	100,000 per VSLA
Thematic area 4: Community infrastructure and disaster management				
Train selected community emergency response teams in Mbuyuni, Ondoni and Silala villages	Design and implement a food for work programme during drought targeting the youth to restore drainage channels, clean culverts, construct gabions and plant trees for erosion and flood control	Construct raised footbridges/culverts at flood-prone crossing points in Mbuyuni, Ondoni and Silala villages	Kilifi County Government, KERRA, NGOs and Private Sector	BOQs required based on numbers and reputable training institutions

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
		Improve road drainage and install culverts to channel floodwater in Ondoni village		
	Develop and operationalize Early Warning Systems (EWS) for floods, drought, pests and diseases (forecasts via chiefs, barazas, media) in Mbuyuni, Ondoni, Silala, and Vwevwesi villages	Equip community emergency response teams (PPEs, first aid kits) in Mbuyuni, Ondoni and Silala villages Construction of a health facility in Silala and Vwevwesi villages	KMD, NDMA Kilifi County Government, Ministry of Interior and National Administration Kenya Red Cross, NGOs and Private Sector	5,000 per day per trainee 70,000 per village for EWS BOQ for equipment and health facility
Thematic area 5: Conflicts resolution (Human Wildlife Conflicts)				
Sensitize Ondoni and Vwevwesi communities on potential conflict hotspots, minimizing human-wildlife interactions and reporting to KWS (hotline) as well as compensation procedures		Installation of solar power electric fence around Kaya Chonyi	KWS, NMK, Kilifi County Government, NGOs, Private Sector	7,000,000
Thematic area 6: Gender and social inclusion to enhance adaptation action				
Design targeted GBV anticipatory actions for drought and floods climate hazards	Develop and implement a Ward GBV anticipatory action plan in collaboration with relevant partners	Establish GBV recovery center (this could be at the county level)	National Government, Kilifi County Government and NGOs, National Government, Kilifi County Government, National Police Service, NGOs, GVRC, COVAW, HAK	1,000,000 for consultancy services Construction of GBV recovery center subject to a detailed Bill of Quantities
Assess the existing early warning systems to produce and transmit reliable and timely information and forecasts for the implementation of GBV	Strengthen gender-responsive governance-inclusion of women, youth and PADs through capacity building across Mbuyuni, Ondoni, Silala, Vwevwesi		Kilifi County Government and NGOs	5,000 per person per day for training

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
anticipatory actions ahead of climate hazards				
Use of existing women associations such as VSLAs to train women and girls on their rights and GBV response strategies, helplines and recovery centers for women and girls as well as the role of men in peaceful conflict resolution	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		Kilifi County Government, Civil Society, NGOs, Private Sector	30,000 per VSLA BOQ for GBV response mechanisms
Organize and hold youth boot camps on guidance and counselling focusing on drugs, antisocial behavior and reproductive health	Promote vaccination of children and youth against diseases such as cholera which could arise and spread on account of climate hazards Support school feeding programmes and scholarships	Construct drug rehabilitation centre targeting the youth (this can be done at the county level)	National Government, NACADA, Kilifi County Government, NGOs, Private Sector	100,000- Boot camp, Vaccination at local medical facilities, BOQ required for drug rehabilitation centre School feeding and scholarships needs assessment
Facilitate PADs to link with organizations that can provide infrastructural assistance such as wheelchairs, specialized training and entrepreneur skills			Kilifi County Government, NGOs, Private Sector	50,000 for meetings

5.4 Resource Mobilisation

Implementation of the Ward Adaptation Plan will rely on a diversified mix of financial, human, and technical resources drawn from county, national, donor, and community-based funding streams. The primary sources include county budget allocations, where relevant departments such as Water, Environment and Natural Resources, Agriculture, Livestock and Blue Economy, and Public Works and Infrastructure will integrate adaptation priorities into their Annual Development Plans and sector budgets. Mwarakaya Ward is projected to receive at least KES 2–3 million annually earmarked for ward-level climate resilience, as required under the Climate Change Act, 2023.Additional financing will be accessed through the Kilifi County Climate Change Fund (CCCF), which channels grants from development partners including the World Bank's FLLoCA programme, supporting locally led adaptation initiatives. At the national level, Mwarakaya Ward will benefit from funds such as the National Drought Emergency Fund and other climate financing mechanisms including the Kenya Adaptation Fund Programme, coordinated through the National Treasury's Climate Finance Unit and implemented in collaboration with agencies like NDMA and KMD.

Development partners and NGOs, including Plan International, WWF, World Vision, Kenya Cereal Enhancement Programme (KCEP), and Botanical Garden Conservation International (BGCI), will continue to support ecosystem restoration, capacity building, and livelihood diversification projects. Community contributions through in-kind labor, local materials, savings groups, Water Resource Users Associations (WRUAs), and small user fees will foster ownership and sustainability. The private sector will be engaged through partnerships for water infrastructure development, crop and livestock insurance, microfinance for smallholder irrigation, digital early warning SMS systems, and corporate social responsibility initiatives. Overall, Mwarakaya Ward expects to mobilize approximately KES 25–35 million over five years from this blended financing framework, ensuring continuity and long-term sustainability of adaptation interventions.

6 Chapter 6: Adaptation Monitoring and Evaluation

6.1 Introduction

Monitoring, Evaluation, and Learning (MEL) are critical components of the Ward Adaptation Plan to ensure that the proposed actions are achieving their intended outcomes and to facilitate continuous improvement. This chapter outlines how Mwarakaya Ward will track progress of adaptation interventions (Monitoring), assess their effectiveness and impact (Evaluation), and capture lessons to inform future climate-resilience planning (Learning). A robust M&E framework not only provides accountability to the community and funders but also enables adaptive management, meaning the Ward Climate Committee can adjust strategies based on what is working or not working on the ground. Given the long-term nature of climate change, M&E will help maintain momentum and demonstrate how resilience is being built incrementally. The approach will utilize participatory methods so that community members are involved in monitoring their own resilience, alongside technical indicators aligned with county and national reporting requirements. The Climate Change Act 2023 reinforces the importance of reporting and transparency in climate actions, and this plan's M&E system will align with those national guidelines while tailoring to local context.

6.2 The Adaptation Theory of Change

The theory of change for Mwarakaya Ward describes the logical linkages between planned interventions and the desired long-term outcomes of climate resilience and sustainable development. It demonstrates how resources and activities will lead to measurable improvements in water security, food production, environmental health, and social well-being.

- **Inputs/Activities:** These include investments such as drilling and equipping boreholes, promoting rainwater harvesting, providing agricultural training, establishing community tree nurseries, implementing flood control infrastructure, and conducting awareness and capacity-building sessions.

- **Outputs:** Tangible short-term results include functional water systems, trained farmers applying climate-smart techniques, community early warning systems established, pest and disease surveillance networks operational, and strengthened local institutions such as the Ward Climate Change Planning Committee (WCCPC).

- **Outcomes:** In the medium term, the community experiences improved access to safe water, increased agricultural productivity and income stability, reduced pest and disease losses, and decreased vulnerability to floods and drought. Gender equity improves through women's leadership roles in adaptation projects and reduced household workloads due to improved access to water and energy-saving technologies.

- **Impact (Goal):** The long-term goal is an adaptive and climate-resilient Mwarakaya community capable of withstanding droughts, floods, and other climate stresses while sustaining livelihoods, protecting ecosystems, and maintaining social cohesion. This includes food-secure households, restored ecosystems, reduced dependency on relief aid, and enhanced capacity of local institutions to plan and manage adaptation sustainably

The theory of change for Mwarakaya can be summarized as: 'IF we strengthen local institutions, promote sustainable water and land management, support climate-resilient agriculture, and build inclusive community capacity, THEN the vulnerability of households to climate hazards will reduce and their adaptive capacity will increase, BECAUSE they will have access to resources, information, and systems that enable proactive and collective action against climate shocks.'

This logical flow will guide M&E. Each level (input -> output -> outcome -> impact) will have specific indicators to check. For instance, training farmers (input) should lead to output of farmers implementing new techniques, which yields outcome of yield stability in drought, contributing to impact of overall food security. If a link is failing (e.g., outputs achieved but expected outcome not observed), it signals a need to investigate and possibly modify the strategy. Importantly, this plan assumes certain enabling conditions (or intermediate outcomes) as part of the theory of change: community engagement remains strong, the local government continues supportive policies, and external shocks beyond climate (like economic downturns or conflicts) do not severely undermine efforts. M&E will also keep an eye on these assumptions. In the theory of change, gender and social inclusion are cross-cutting; by involving women and youth in decision-making (output), we expect more equitable resilience building (outcome), contributing to a more cohesive and adaptable society (impact). Likewise, capacity building of institutions (output) yields better coordination and resource access (outcome), leading to sustainable implementation (impact). By explicitly stating this theory, all stakeholders have a common understanding of what success looks like and how we plan to get there, which will inform what we monitor and evaluate.

6.3 Performance Indicators for Adaptation

Based on the theory of change, a set of Performance Indicators will be used to measure progress and effectiveness. These indicators will be both quantitative and qualitative, and they align with national adaptation M&E frameworks where possible (Kenya's NAP/NCCAP indicators) as well as community-defined measures of success. The performance indicators are structured at various levels:

Input/Output Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Number of household rainwater harvesting systems installed (gutters + tanks).
   - Number of schools equipped with high-capacity rainwater harvesting systems.
   - Number of water pans constructed
   - Number of manual and solar-powered pumps installed on existing wells.
   - Number of women, youth, and PADs using solarised/manual hand pumps.
   - Number and percentage of women, youth, and PADs in water management committees.
   - Amount of budget allocated and utilized for water infrastructure.
   - Number of boreholes drilled, equipped, and solarised.
   - Number of boreholes fully operational and supplying water year-round.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

   - Number of farmers trained on Resilience Design techniques.
   - Number of farmers trained on CSA practices (land prep, planting, pest control, post-harvest).
   - Acreage under drought-tolerant crops supported with certified seeds.
   - Number of Farmer Field Schools established and functional.
   - Number of demonstration farms established by women and youth lead farmers.
   - Number of farmers receiving inputs (certified seeds, fertilizers, pesticides).
   - Number of households supported with organic manure production and composting.
   - Number of farmers adopting terracing, zai pits, swales, and other soil/water conservation measures.
   - Number of livestock fodder seed banks and pasture plots established.
   - Number of households receiving improved livestock breeds (Gala goats, dairy cows).

- Number of farmers sensitised on crop/livestock insurance. Number of households receiving indigenous/fruit seedlings.
- Number of community seedling distribution events held.
- Number of on-farm tree planting sessions conducted.
- Number of mangrove nurseries supported and maintained.
- Number of acres of mangrove restored through enrichment planting.
- Number of community trainings on mangrove protection and sustainable harvesting.
- Number of NBS windbreaks (Neem, Casuarina) planted across all villages.
- Number of environmental conservation awareness sessions held.
- Number of households adopting sustainable charcoal alternatives or clean energy.
- Number of households supported with grain storage equipment (hermetic bags, grain banks).
- Number of village pest and disease surveillance teams established and trained.
- Budget allocated and utilized for restoration activities.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION
   - Number of women and youth trained in alternative livelihoods (eco-tourism, beekeeping, poultry).
   - Number of modern beehives, incubators, or eco-tourism boats distributed.
   - Number of households supported with livelihood start-up kits.
   - Number of community groups formed for value addition and market linkages.
   - Number of training sessions on business development, value addition, and financial literacy.
   - Number of social protection initiatives implemented (food distribution, cash transfer mobilizations).
   - Number of school feeding programmes supported.
   - Budget allocations for livelihood diversification (construction of cultural center in Kay Chonyi), implemented.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

   - Number of culverts reconstructed or improved with proper drainage.
   - Number of flood-prone crossing points fitted with raised footbridges.
   - Number of youth mobilized in food-for-work programmes for drainage, gabions, and erosion control.
   - Number of villages with functional flood Early Warning Systems (EWS) linked to KMD.
   - Number of community dissemination channels established (chief barazas, noticeboards, SMS).
   - Number of community emergency response teams trained.
   - Number of community emergency response teams equipped (boats, PPE, first-aid kits).
   - Number of designated disaster assembly points established and labelled.
   - Budget allocated and utilized for infrastructure upgrades (roads, culverts, drainage).

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)
   - Number of community sensitization sessions on human–wildlife conflict hotspots.
   - Number of households supported with HWC reporting mechanisms (hotline awareness).
   - Number of kilometers of solar-powered fencing installed in hotspot areas (e.g. Ondoni and Vwevwesi)
   - Number of community trainings on safe behavior in baboon, monkeys and snake-risk zones.

- Number of reported HWC cases logged and submitted to KWS for follow-up.
- Funding allocated and utilized for HWC mitigation (fencing, awareness, surveillance).

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS
   - Number of gender-responsive governance trainings conducted.
   - Number and percentage of women, youth, and PADs participating in ward committees.
   - Number of VSLAs supported with GBV awareness and rights training.
   - Number of GBV anticipatory action sessions conducted for drought, floods, and strong winds.
   - Number of GBV response kits (first-aid kits, rape kits) distributed to health facilities.
   - Number of early warning systems integrating GBV alert messaging.
   - Number of PADs supported with assistive devices, rehabilitation linkages, or livelihood equipment.
   - Number of youths bootcamps conducted on drugs, behaviour change, and reproductive health.
   - Number of awareness campaigns on GBV helplines and safe reporting mechanisms.
   - Number of meetings with PAD support organizations facilitated.

Outcome Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Reduction in average time spent collecting water (minutes/day), disaggregated by women, youth, and PADs.
   - Increase in household water storage capacity (litres/HH).
   - Percentage of water facilities functional year-round.
   - Increase in number/percentage of women, youth, and PADs with independent access to water sources (boreholes, solarised pumps, kiosks).
   - Increased reliability of water supply during drought periods (days/months of service continuity).

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS
   - Stability of crop yields over 3–5 seasons among supported farmers.
   - Improvement in soil health indicators, including:
   - Increase in soil organic matter (%)
   - Increase in soil moisture retention capacity
   - Improvement in soil nutrient status (N, P, K)
   - Improved aggregate stability and soil structure
   - Percentage of farmers adopting integrated soil and water conservation (terracing, zai pits, swales).
   - Increase in acreage under diversified and climate-resilient farm systems.
   - Increase in household-level adoption of post-harvest management technologies (grain banks, hermetic bags).
   - Percentage of women, youth, and PADs participating in CSA and Resilience Design training cycles
   - Increase in on-farm and community tree cover (%) over baseline.
   - Total acreage restored areas in Kaya Chonyi.
   - Reduction in incidents of illegal harvesting, charcoal burning and sand harvesting
   - Increase in households adopting windbreak and shelterbelt systems.
   - Increased community participation in environmental conservation and sustainable harvesting practices.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION
   - Percentage increase in household income from diversified livelihoods (eco-tourism, poultry, beekeeping, seaweed value addition).
   - Increase in the number of households with multiple income streams across seasons.
   - Percentage of women, youth, and PADs accessing new livelihood opportunities.
   - Increased financial literacy and business skills among trained groups.
   - Increase in households benefiting from school feeding, cash transfers, or food-for-work schemes.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT
   - Reduction in households exposed to flood-prone or erosion hotspots.
   - Percentage of villages receiving timely Early Warning System (EWS) alerts.
   - Community response rate to EWS alerts (attendance at assembly points, evacuation compliance).
   - Reduction in flood-related damages to dwellings and access roads.
   - Level of preparedness and readiness of community emergency response teams (measured via drills).

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)
   - Reduction in frequency of human–wildlife conflict incidents (baboon raids).
   - Increase in the number of households protected by solar-powered fences.
   - Increase in community capacity to report and respond to HWC events (via KWS hotline).
   - Increased community compliance with safe-behaviour protocols in HWC hotspots.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS
   - Increase in the number and proportion of women, youth, and PADs participating in ward-level governance and adaptation committees.
   - Reduction in drudgery for women, youth, and PADs due to improved water access and resilient infrastructure.
   - Percentage increase in women's and PADs' access to credit, extension services, and productive resources.
   - Increase in women's ownership of productive assets (livestock, seaweed farms, equipment).
   - Improvement in community knowledge and awareness of GBV risk mitigation during climate hazards.
   - Increased utilisation of GBV response services (safe shelters, first-aid kits, referral pathways).

Impact (Goal) Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Percentage of the ward population with year-round access to safe and reliable water.
   - Increase in average litres of water available per person per day across villages.
   - Long-term reduction in water scarcity–related vulnerabilities for women, youth, and PADs.
   - Sustained decrease in waterborne diseases due to improved water quality and availability.
   - Community-wide resilience to prolonged drought through diversified water sources and storage systems.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOOD

- Sustained increase in crop yields above baseline across 3–5 year periods.
- Long-term reduction in seasonal yield fluctuations linked to droughts, floods, and pest outbreaks.
- Measurable improvements in soil health at landscape level, including:
- Higher soil organic carbon (SOC)
- Increased vegetation cover (NDVI index)
- Reduced erosion and sediment loss
- Enhanced household food security across all seasons.
- Increased biodiversity within integrated farming systems (trees, fodder, pollinator presence).

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION
   - Sustained increases in household incomes from diversified livelihoods (beekeeping, eco-tourism, and poultry).
   - Improved household resilience to income shocks from climate hazards.
   - Decreased poverty levels due to stable and multi-stream livelihoods.
   - Increased school attendance and nutritional outcomes for children due to social protection schemes.
   - Greater financial stability among women, youth, and PAD households.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMNT

- Significant reduction in climate-related damage to homes, roads, and community facilities.
- Communities demonstrating effective disaster preparedness and response during floods and storms.
- Long-term reduction in displacement, injuries, and asset losses during climate hazards.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)
   - Long-term reduction in livestock losses, and crop destruction from wildlife.
   - Improved coexistence between communities and wildlife due to effective mitigation (fencing, reporting systems).
   - Reduced economic losses from HWC-related disruptions.
   - Increased safety and mobility for women, children, and PADs in high-risk zones.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS
   - Percentage increase in women's, youth's, and PADs' participation in climate governance and local decision-making structures.
   - Strengthened continuity of SRHR, maternal health, and GBV response services during climate hazards.
   - Increased leadership roles held by women, youth, and PADs in adaptation implementation.
   - Growth in the number of marginalised and vulnerable groups leading or influencing climate adaptation actions.
   - Reduction in GBV risks during droughts, floods, and strong winds due to anticipatory

of data collection, and responsible person.

For each indicator, we will establish baselines (where we are starting from, using historical data or initial surveys) and set targets for 1-year, 3-year, 5-year intervals. For example: baseline of 0 operational boreholes, target 3 by 2025; baseline average maize yield 8 bags/acre in normal

year and 2 in drought year, target to achieve 5 in a drought year with adaptation by 2027; baseline women fetching water 5 km, target <1 km by 2024, etc.

Data for these indicators will come from various sources: routine project monitoring (engineer reports for infrastructure, training attendance sheets), surveys and focus groups for outcomes (e.g., seasonal household surveys to gauge water access, yields, or time saved), and existing data (health clinic records for disease incidence, meteorological records for hazard frequency). All indicators will be disaggregated by gender where applicable to monitor differential benefits. Also, where feasible, by village to see if some areas lag behind. Measuring adaptation success is complex because it's about avoided losses and capacity built, but these indicators together provide a robust picture. The Ward Committee will use a simple M&E matrix listing each indicator, baseline, target, data source, frequency of data collection, and responsible person.

6.4 Adaptation Reporting and Learning

Reporting: Mwarakaya Ward will establish a clear reporting system to ensure information flows to all stakeholders i.e., community members, county authorities, and funding partners. Internally, the WCCPC will prepare quarterly progress reports that document activities done, outputs achieved, challenges encountered, and expenditures made. These will be shared in ward public barazas for transparency (so the community hears what's been done and can give feedback) and submitted to the County Climate Change Unit. Annually, a more comprehensive Ward Adaptation Report will be produced, summarizing the year's progress against the plan's targets and indicators. This annual report can feed into the county's climate change reporting which is required by the Climate Change Act, ensuring Mwarakaya Ward's experiences contribute to county and national adaptation tracking.

The annual report will highlight case studies or success stories (e.g., "Farmer groups in Ondoni doubled their harvest despite the 2025 drought by using new techniques a success attributed to our adaptation training"). It will also not shy away from discussing failures or shortfalls, as these provide learning.

At community level, participatory monitoring meetings will be held. For instance, after each rainy season, community members can gather to discuss: did the drainage improvements prevent flooding? Did the new seeds perform better in this season? Such qualitative feedback will be recorded and included in reports.

The Ward Administrator and WCCPC Chair will also report in county forums (e.g., County Adaptation Planning Committee meetings) on progress, championing their ward and possibly leveraging more support by showcasing results.

Learning and Reflective Review: Adaptation is a long journey with uncertainty, so fostering a culture of learning is vital. Twice during the implementation period (e.g., mid-term in year 3, and end-term in year 5), the ward will conduct an outcome evaluation and learning workshop. This involves bringing together community representatives, implementing partners, and possibly external evaluators to systematically review what outcomes have been achieved and why. Key questions: Which interventions are most effective? Are there unintended impacts? What obstacles are reducing effectiveness (maybe maintenance issues, or cultural barriers to adoption)? For example, maybe the uptake of an improved cookstove program is low – through reflection the group might learn women prefer a different design, or fuel is an issue, leading to adjusting that action. These learning workshops will use tools like community scorecards, ranking of interventions by community benefit, and problem tree analysis for any persistent challenges (similar to the PCVA style but now focusing on project performance).

Lessons learned will be documented and used to refine the adaptation plan. The Ward Climate Committee is empowered to tweak or reprioritize actions based on evidence. For instance, if a particular crop variety fails, they will note it and try an alternative next season.

Knowledge Sharing: Mwarakaya Ward will participate in peer-to-peer learning exchanges with neighboring wards and other counties. These exchanges, facilitated by the County Climate Change Unit, will allow sharing of practical experiences such as effective forest restoration techniques, community-based early warning dissemination, and VSLA success stories.

Feedback to Policy: Through reporting channels, the ward's experiences may also inform higher level policies. For example, if the ward finds that insurance uptake is low because of certain policy constraints, this feedback can go to the national level via NDMA or the National Climate Change Council to refine such programs. The 2023 Act calls for progressive improvement of climate response, our ward's data and stories become part of that national learning loop.

Sustainability of M&E: The sustainability of M&E will be anchored in local ownership and capacity building. The WCCPC will appoint an M&E focal person or subcommittee to coordinate data collection, analysis, and reporting. Monitoring tasks will be integrated into existing community structures such as Water User Associations, Farmer Field Schools, and Community Forest Associations. Training on basic data management, participatory evaluation, and digital record-keeping will be provided to ensure long-term continuity.

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9 Annexure

1. Priority tree and fruit species

Scientific Name	Common Name
Azadirachta indica	Neem
Afzelia quanzensis	Pod Mahogany / Mkongo
Casuarina equisetifolia	Whistling Pine / Casuarina
Milicia excelsa	African Teak / Iroko / Mvule
Gmelina arborea	Gmelina / White Teak
Tectona grandis	Teak
Grevillea robusta	Silk Oak / Grevillea
Senna spp.	Senna / Cassia
Moringa oleifera	Moringa / Drumstick Tree
Azalea spp.	Azalea
Citrus sinensis	Sweet Orange
Anacardium occidentale	Cashew
Mangifera indica	Mango
Cocos nucifera	Coconut
Carica papaya	Papaya